

Mathematics of Machine Learning

Over-Fitting



Overfitting, regularisation, feature selection

Outline

1. **Over-fitting?**
2. Controlling Complexity
3. Hidden structure
4. Regularisation



Over-fitting

- Complex machine can **over-fit**

over-fitting: fitting the training data well at the cost of getting poorer generalisation performance

- Three red cars. . .
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Binary Classification Task for You



Class 1



Class 2

Which Category?

- Which category does the following image belong to?



Spurious Rules

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- It will find a rule that does this, but not necessary the rule you had in mind
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All Binary Functions

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$$\mathbf{x}_1 = 100 \quad y_1 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_2 = 010 \quad y_2 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_3 = 110 \quad y_3 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_4 = 001 \quad y_4 = \begin{cases} 0 \\ 1 \end{cases}$$

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- Would an MLP that could approximate any function be useful?
- **Absolutely not!**

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Controlling Complexity

- Infinitely flexible machine don't generalise
- Machine learning only works because there is some structure in the data
- A successful machine should capture this structure
- Even deep learning machines with millions or billions of parameters only work because they successfully capture the structure of images or text
- Different learning machines have different performance on different problems because the data has different structure

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Training Examples

- As we increase the number of training examples, we make it hard to find a spurious rule
- Bigger data sets allow us to use more complicated machines
- Part of the success of deep learning is because they use huge training sets
- (Labelled) data is often expensive to collect so we sometimes have no choice but to use a small training set
- One of the limitations of using deep learning comes because we often have limited data

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- In images we believe the identity of an object is invariant to translation and scaling
- The success of *convolutional neural networks* (CNNs) in deep learning is in large part because the convolutions respect translational invariance

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Preprocessing

- Structure might often be obscure to the learning machine
- If we are trying to predict the spread of disease then a list of place names might be a lot less useful than their coordinates
- Imposing an ordering on an unordered set might **not** be useful

$\{ \text{"blue"} : 0, \text{"brown"} : 1, \text{"green"} : 2, \text{"black"} : 3 \}$

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- One view of deep learning is that each layer (particularly in CNNs) acts as a preprocessor
- That is, it finds filters that captures features salient to the problem being tackled
- For both images and texts we expect salient features to be spatially localised
- The deep structure allows ever more complicated features to be captured—that is, we can find spatially localised features on different scales
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- A very successful strategy is to try many different machine learning techniques and choose the best
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
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- Fine tuning hyper-parameter is important and almost always required (true in SVMs, MLP, deep learning)

Measuring Generalisation Performance

- Recall, we want to predict **unseen** data
- **You cannot use data that you have trained on!**
- Need to split your data up into training and validation set
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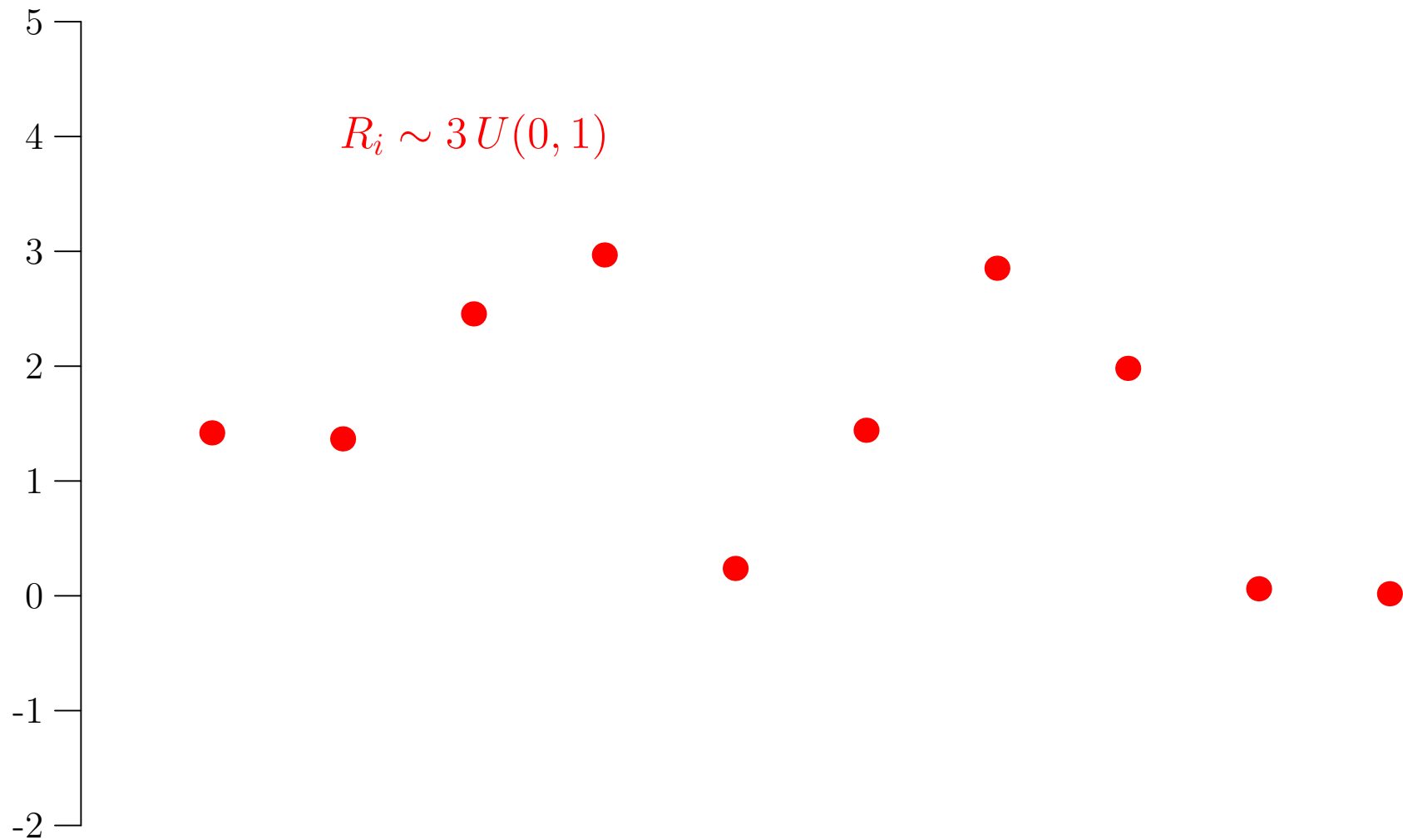
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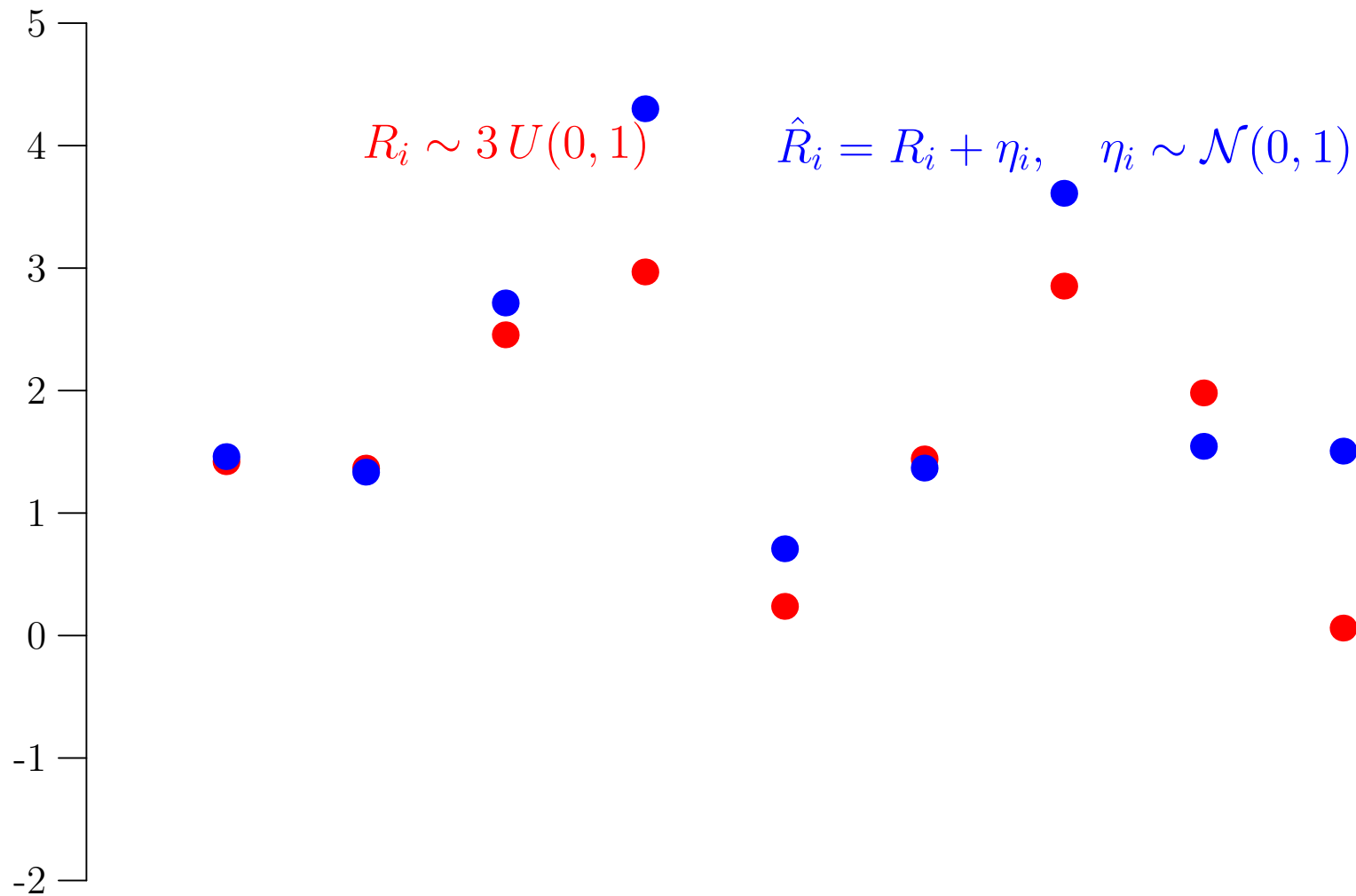
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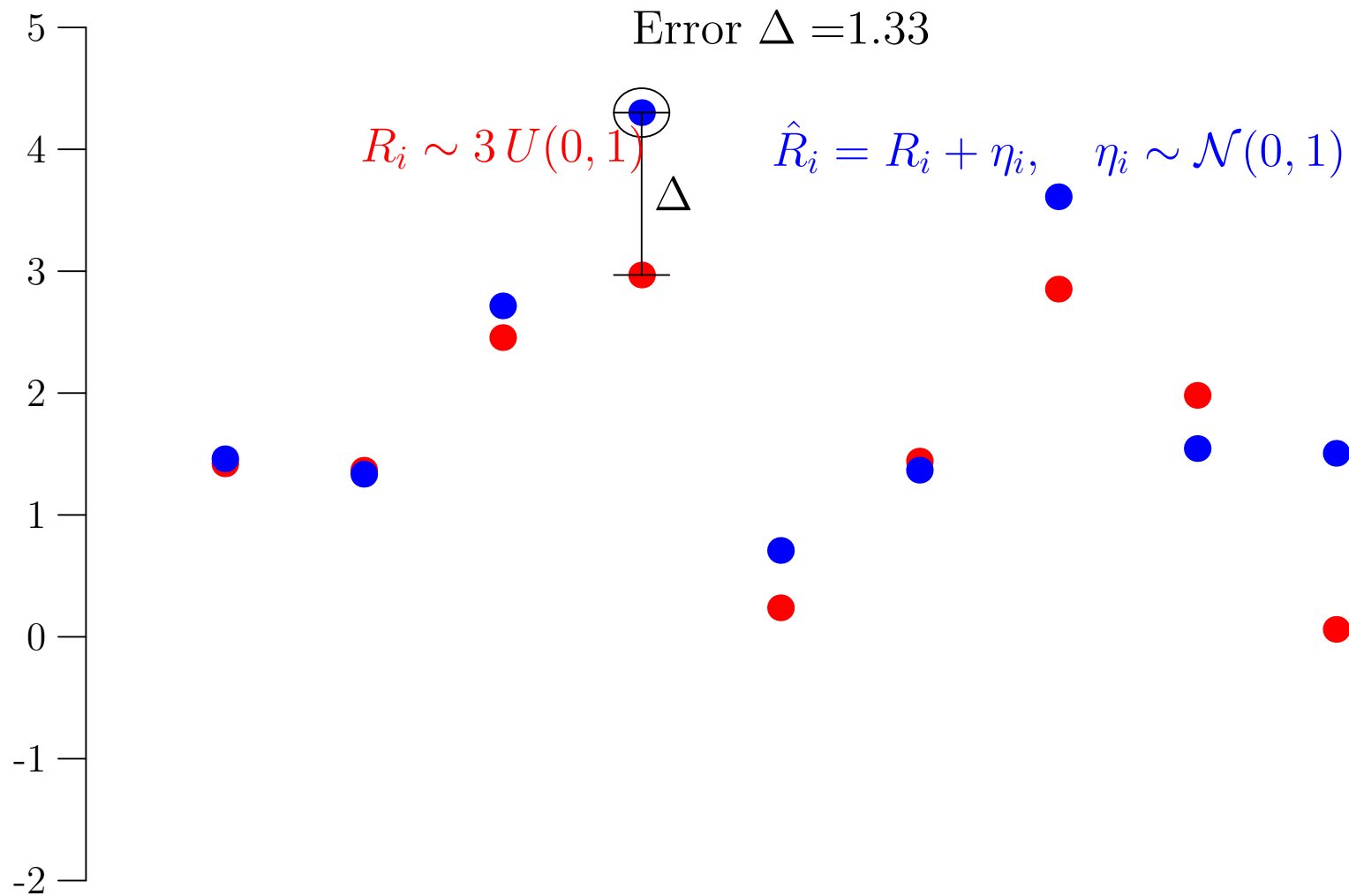
The Overfitting Game



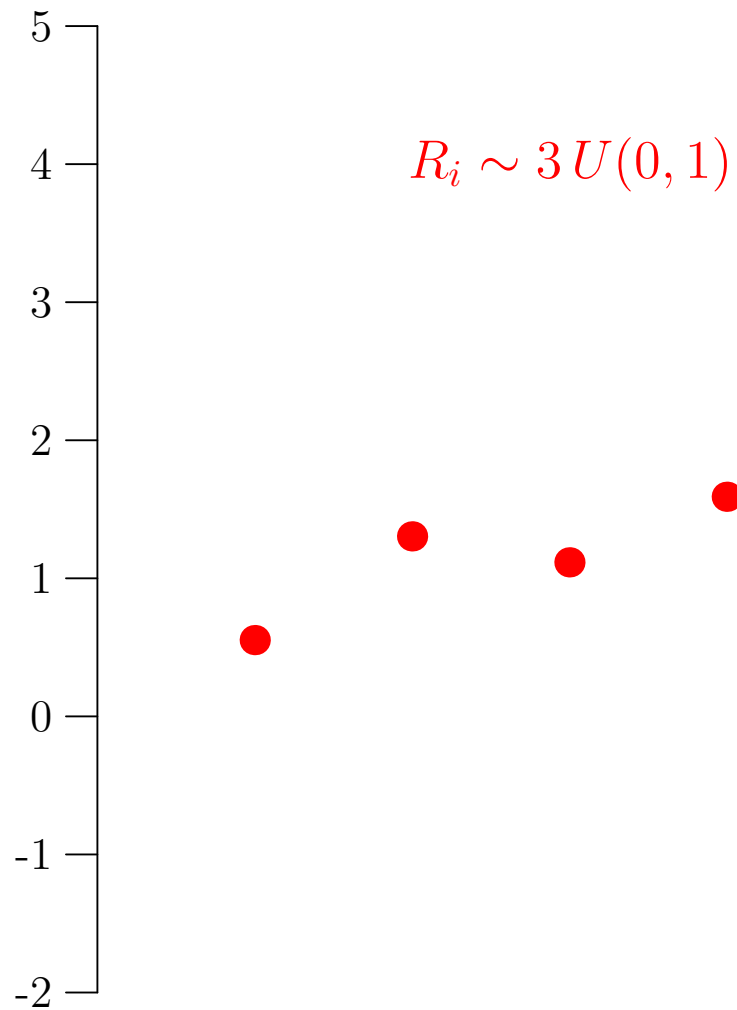
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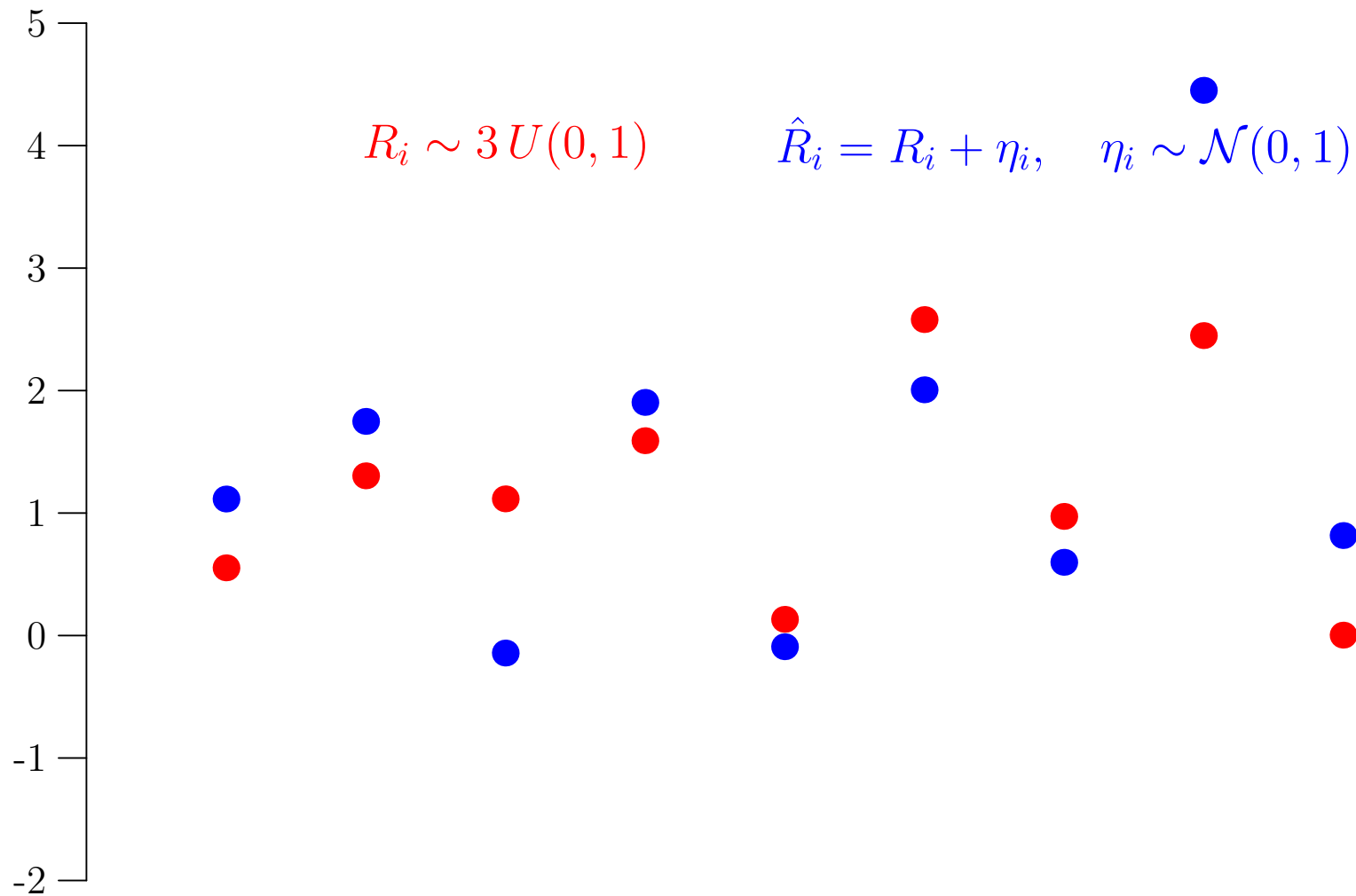
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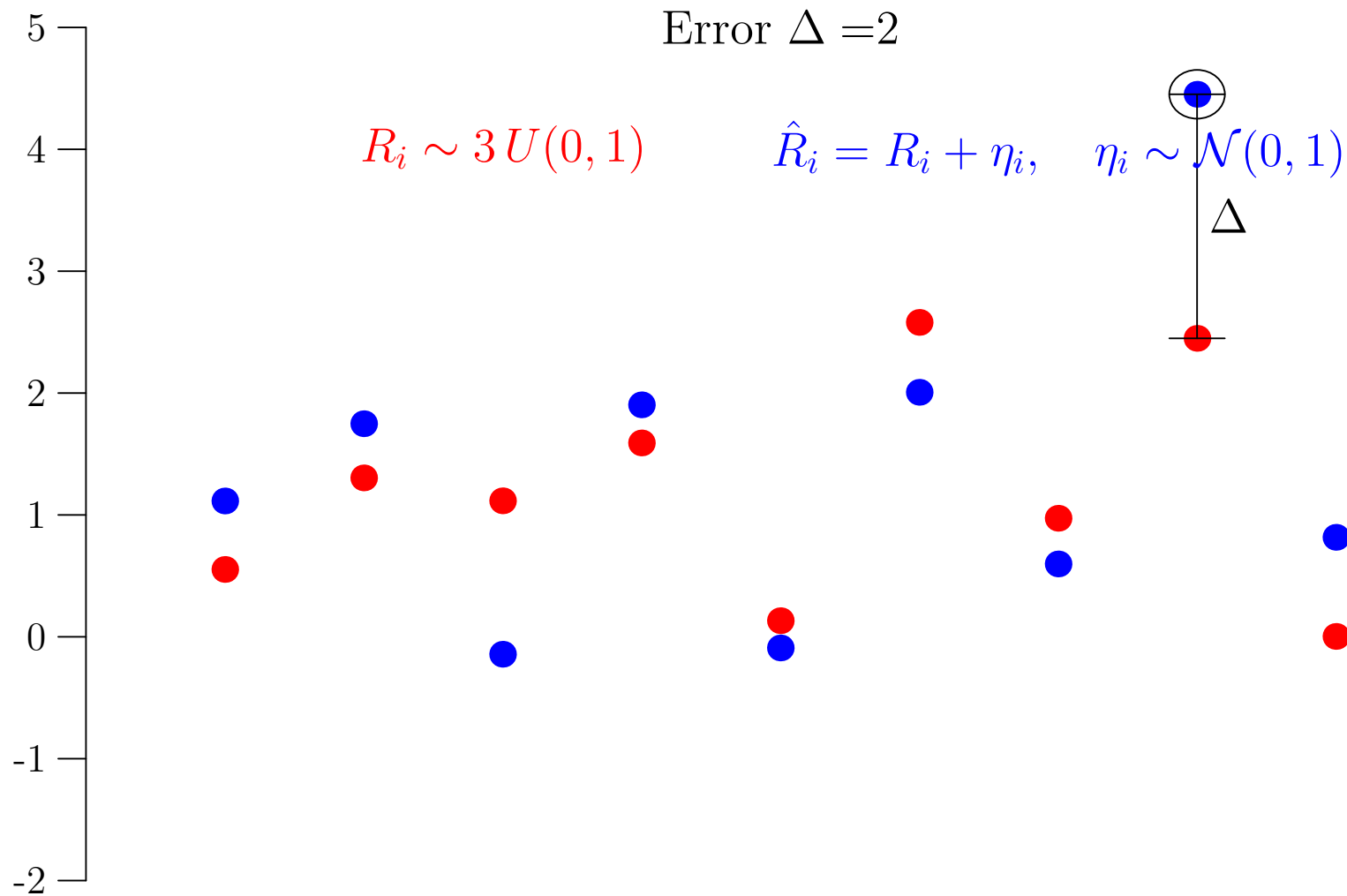
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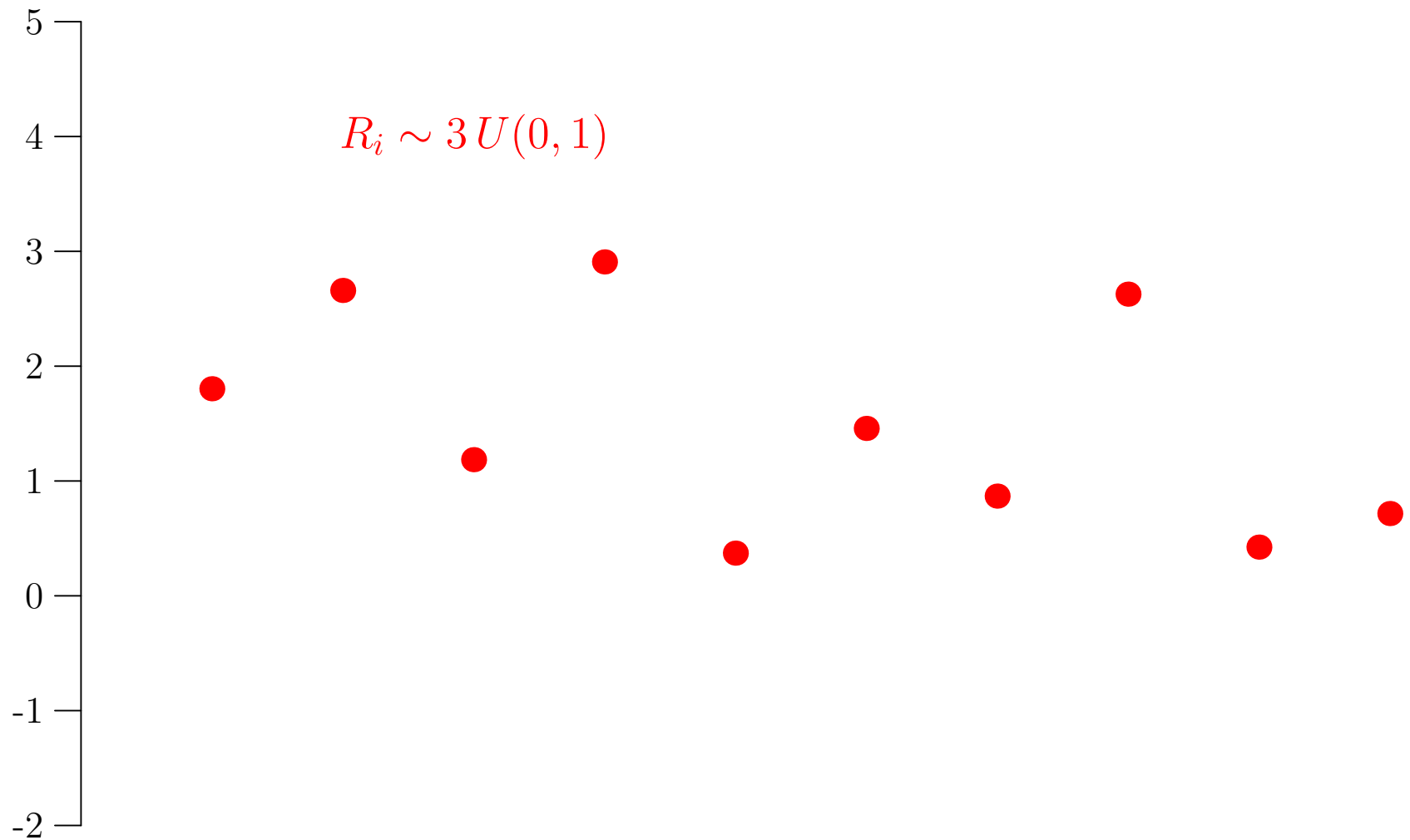
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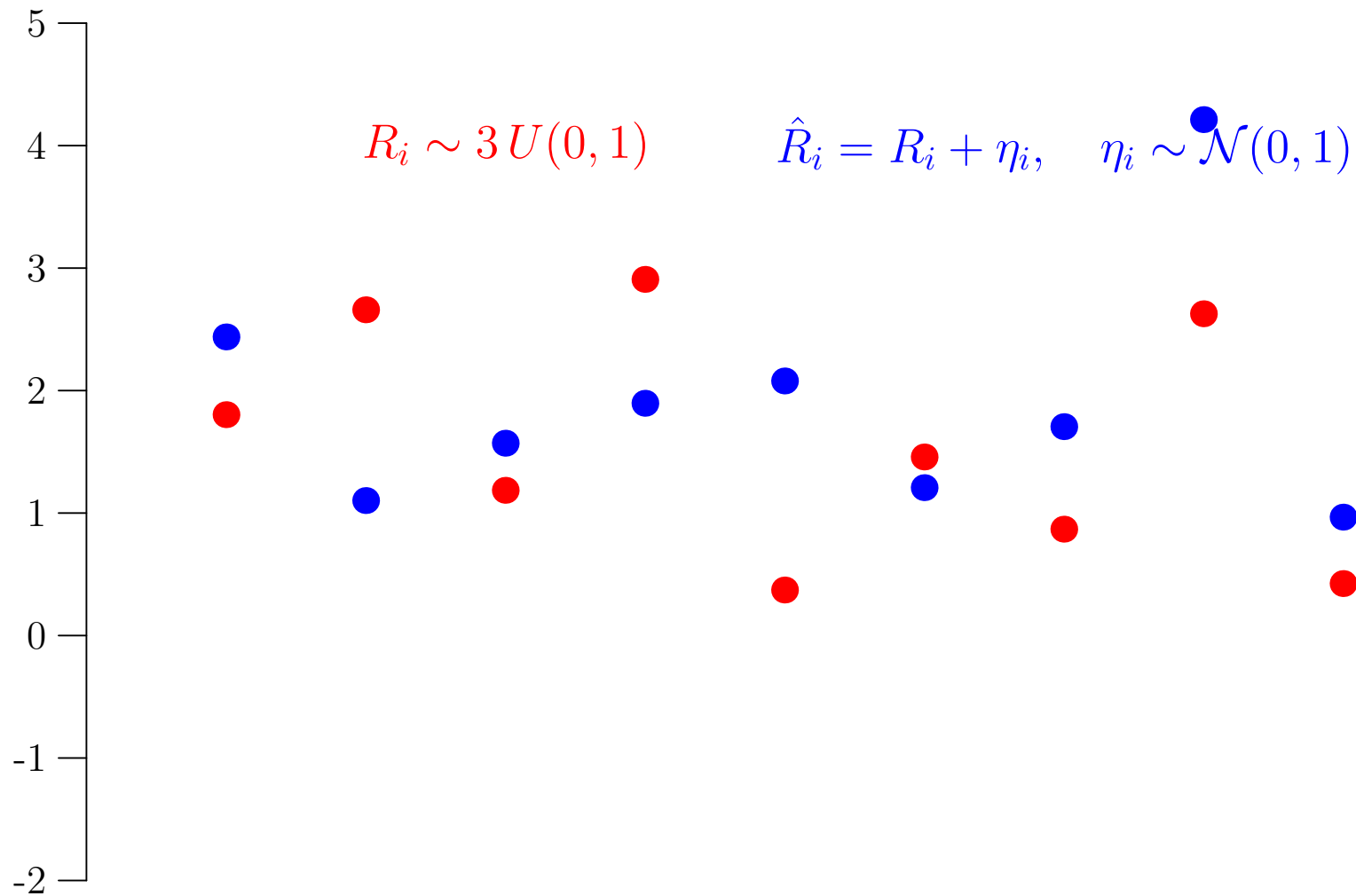
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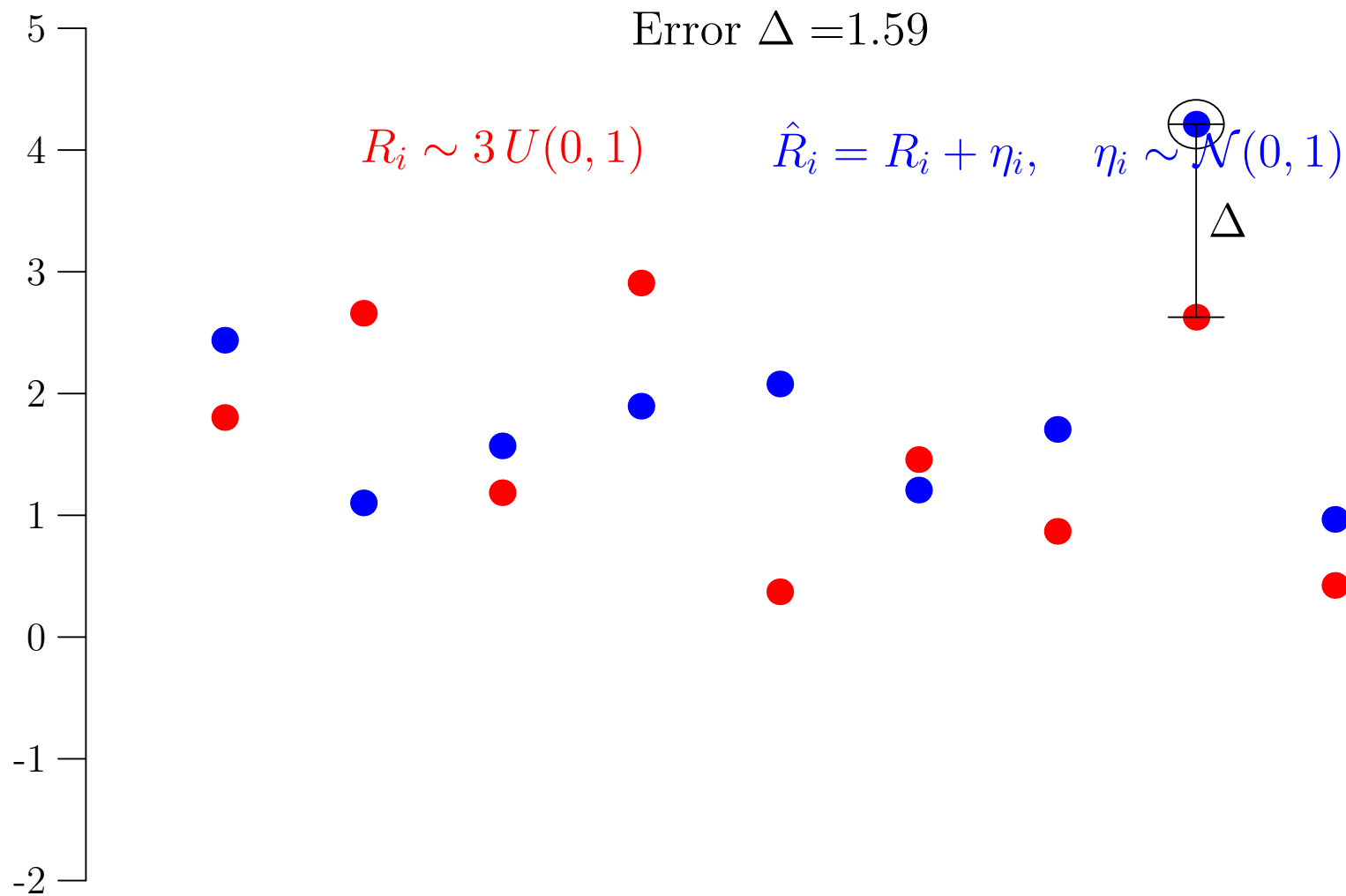
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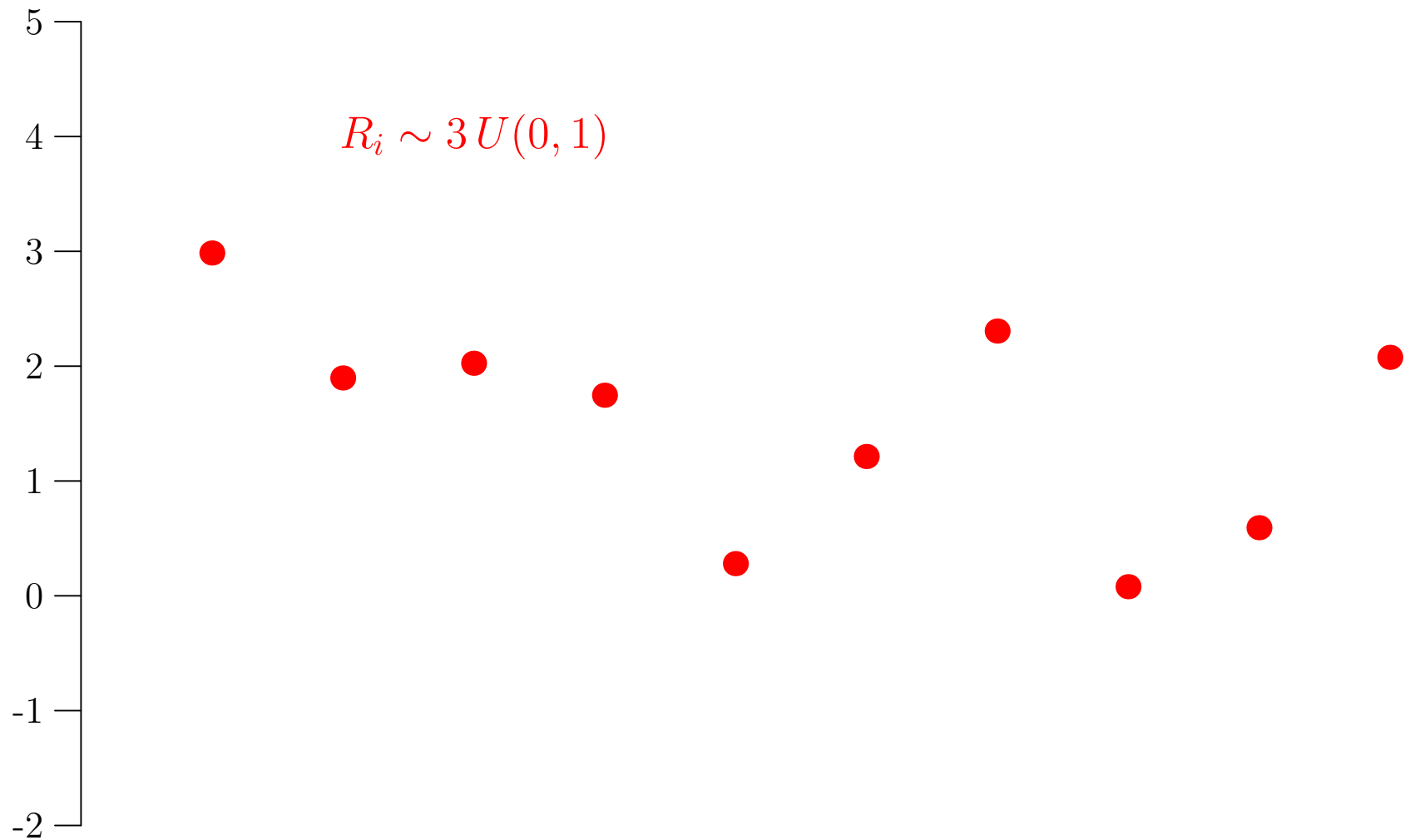
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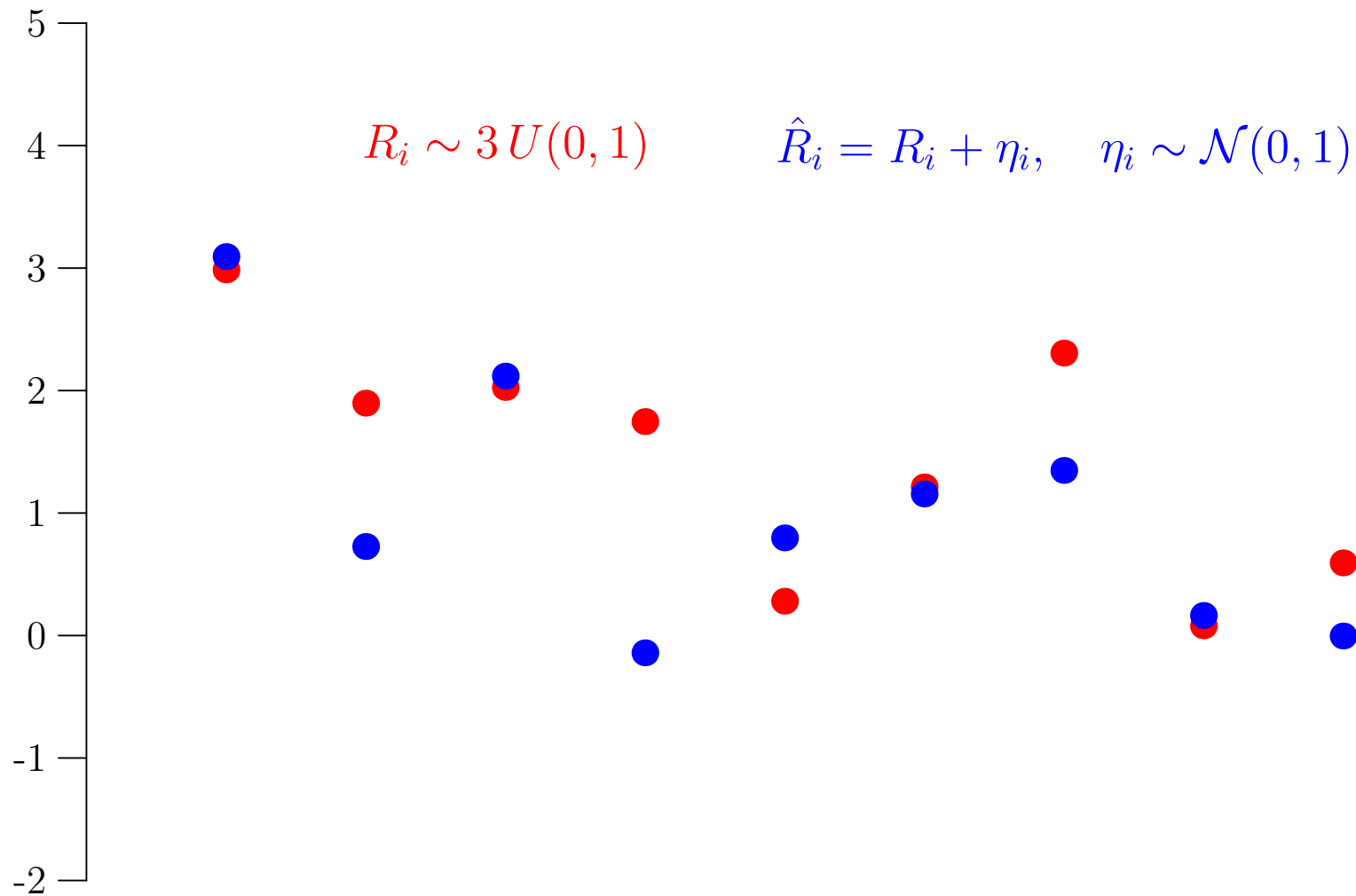
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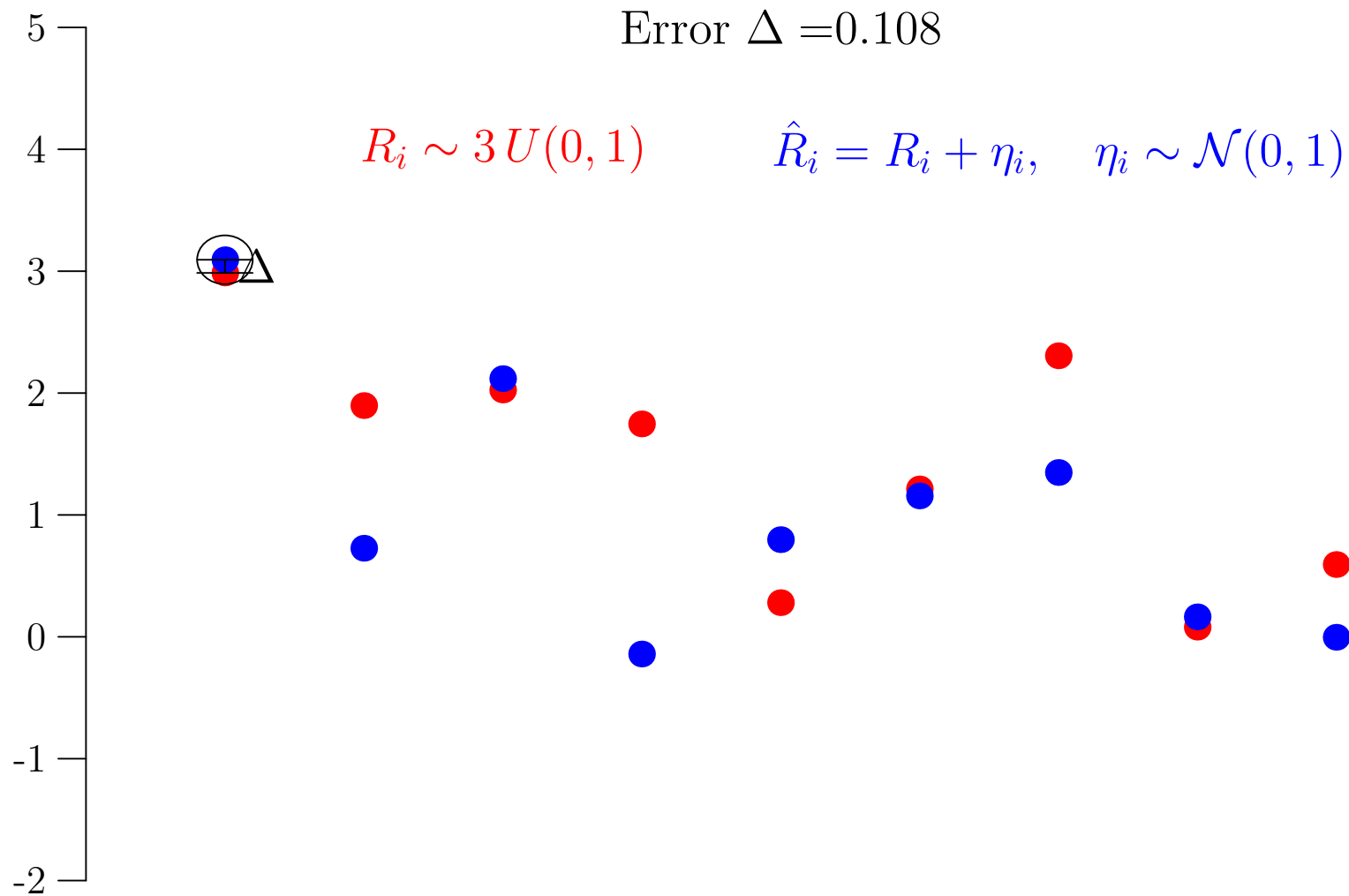
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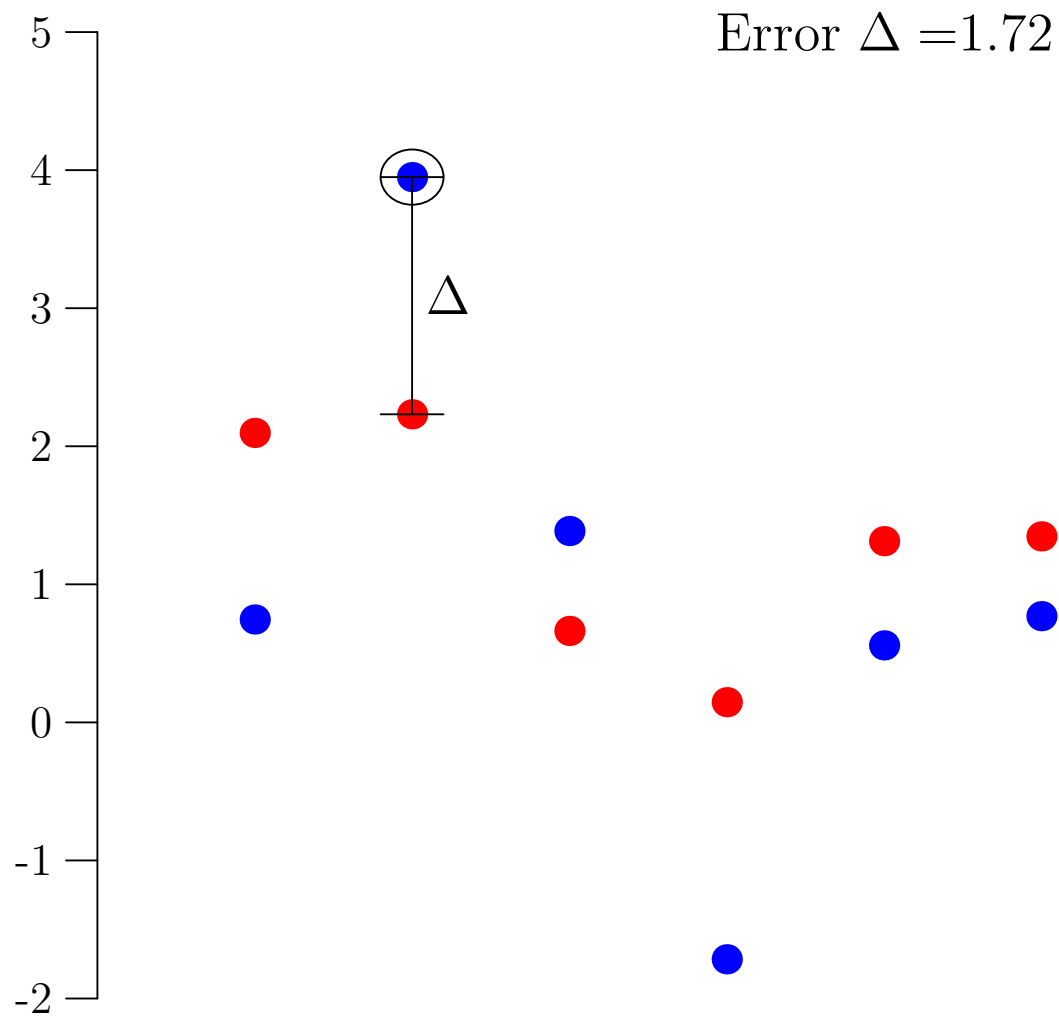
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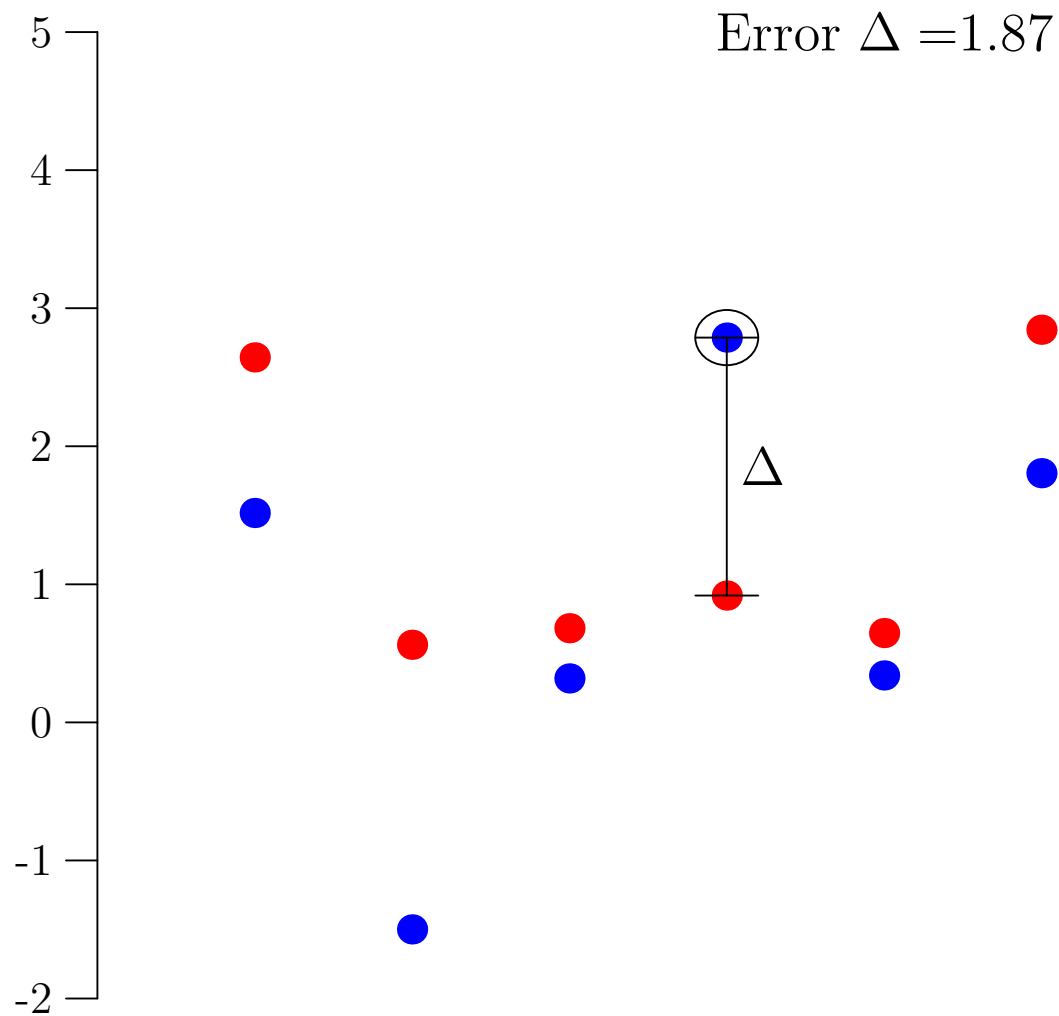
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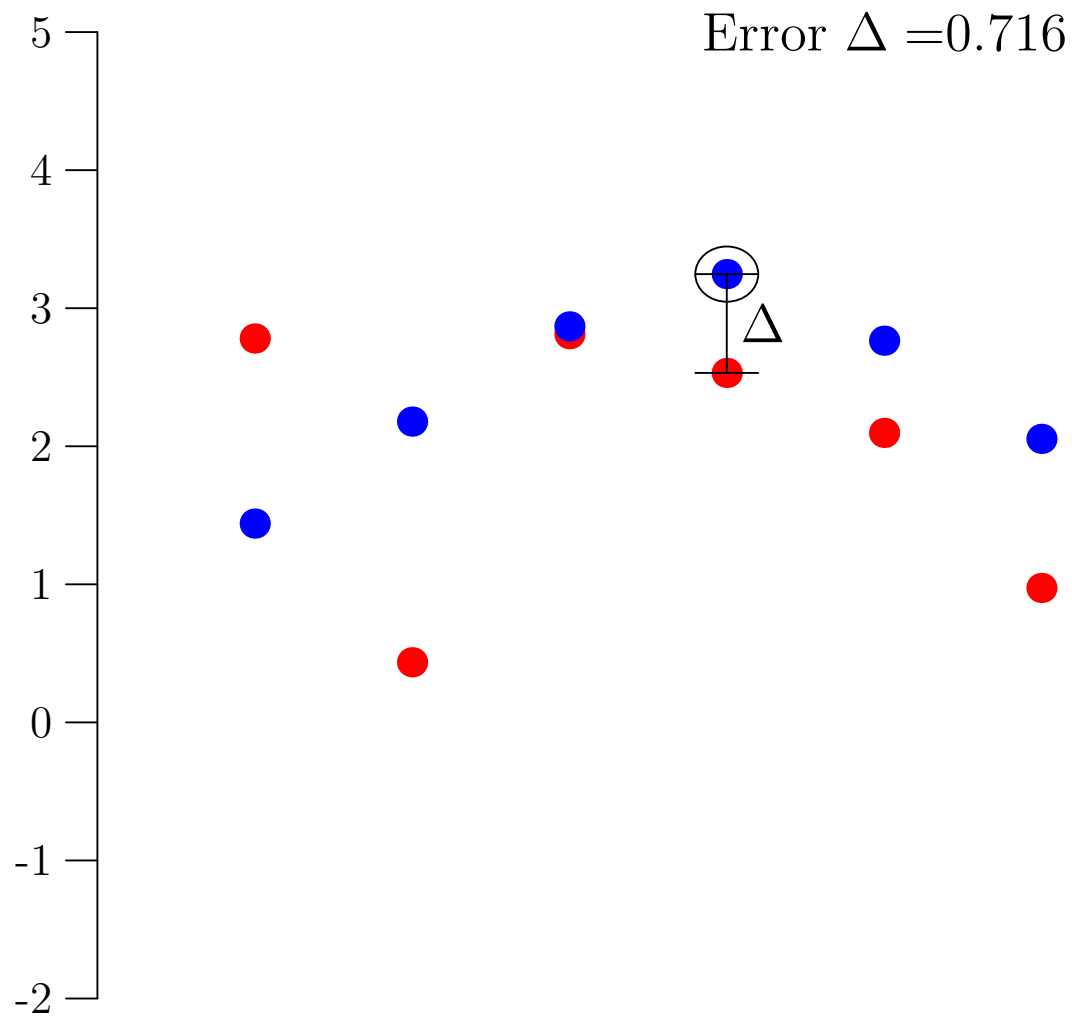
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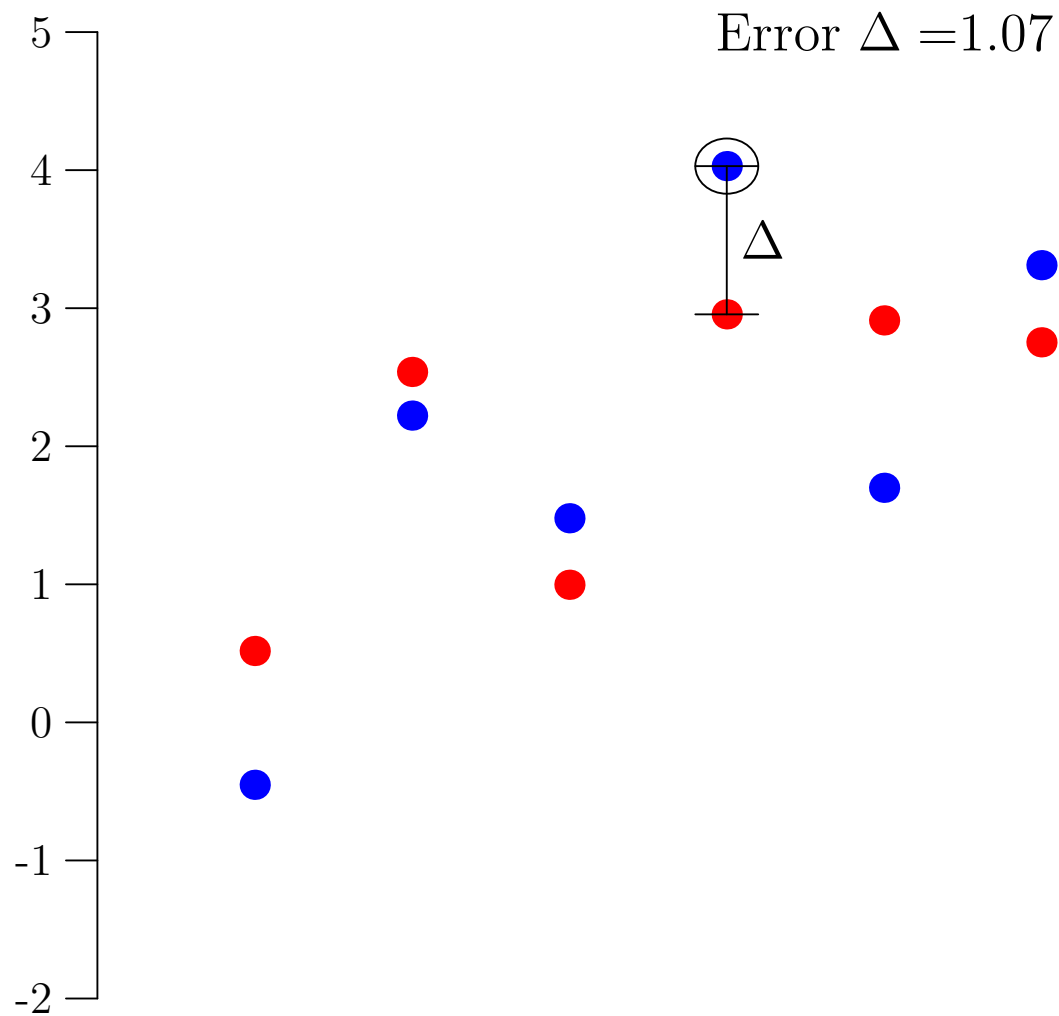
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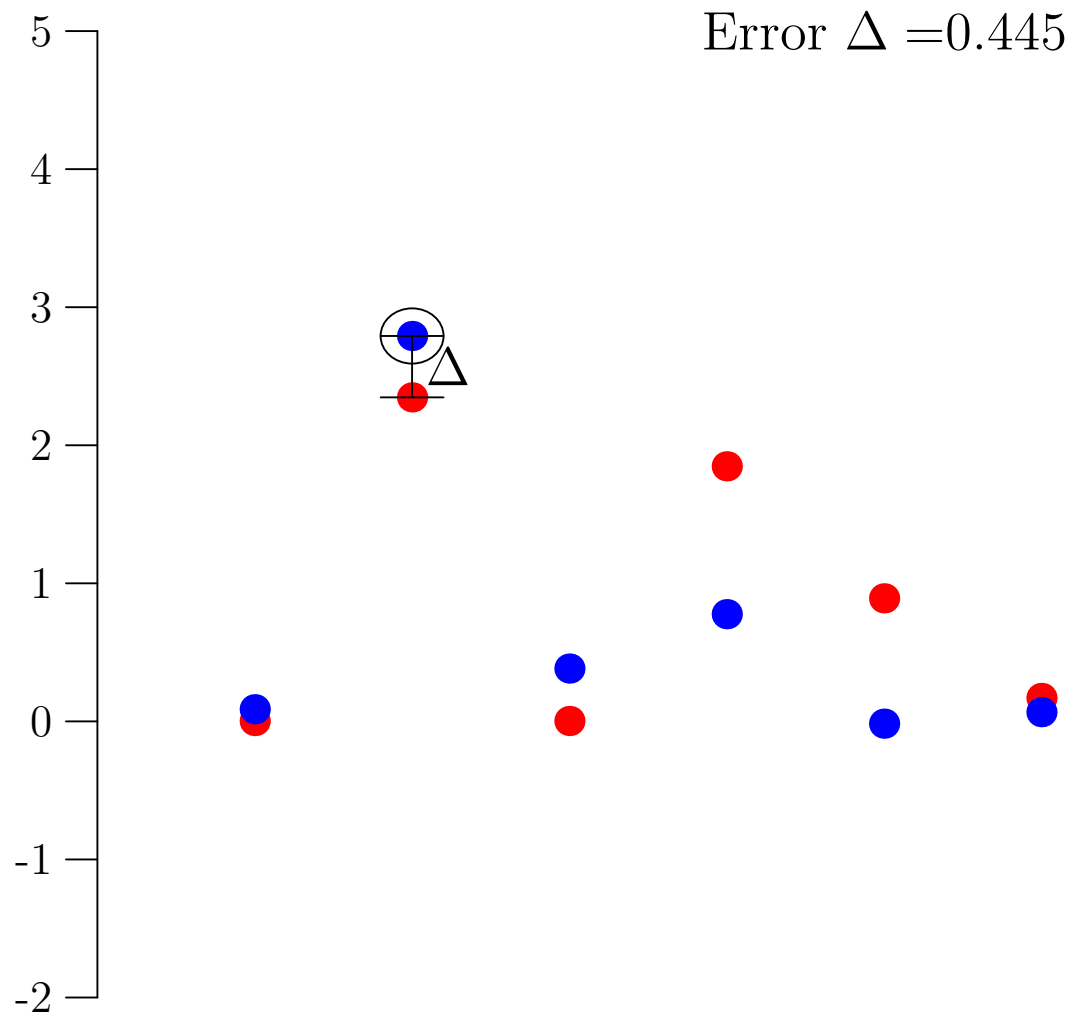
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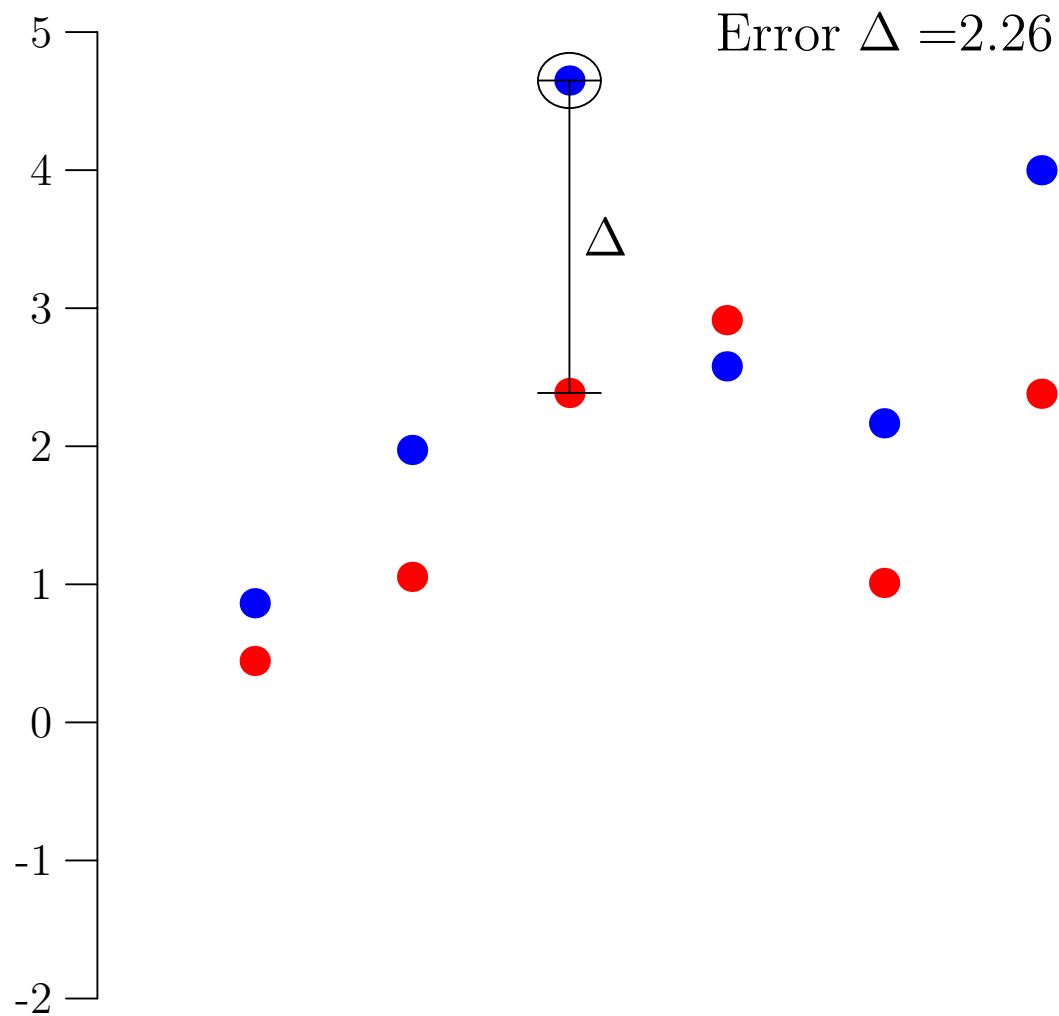
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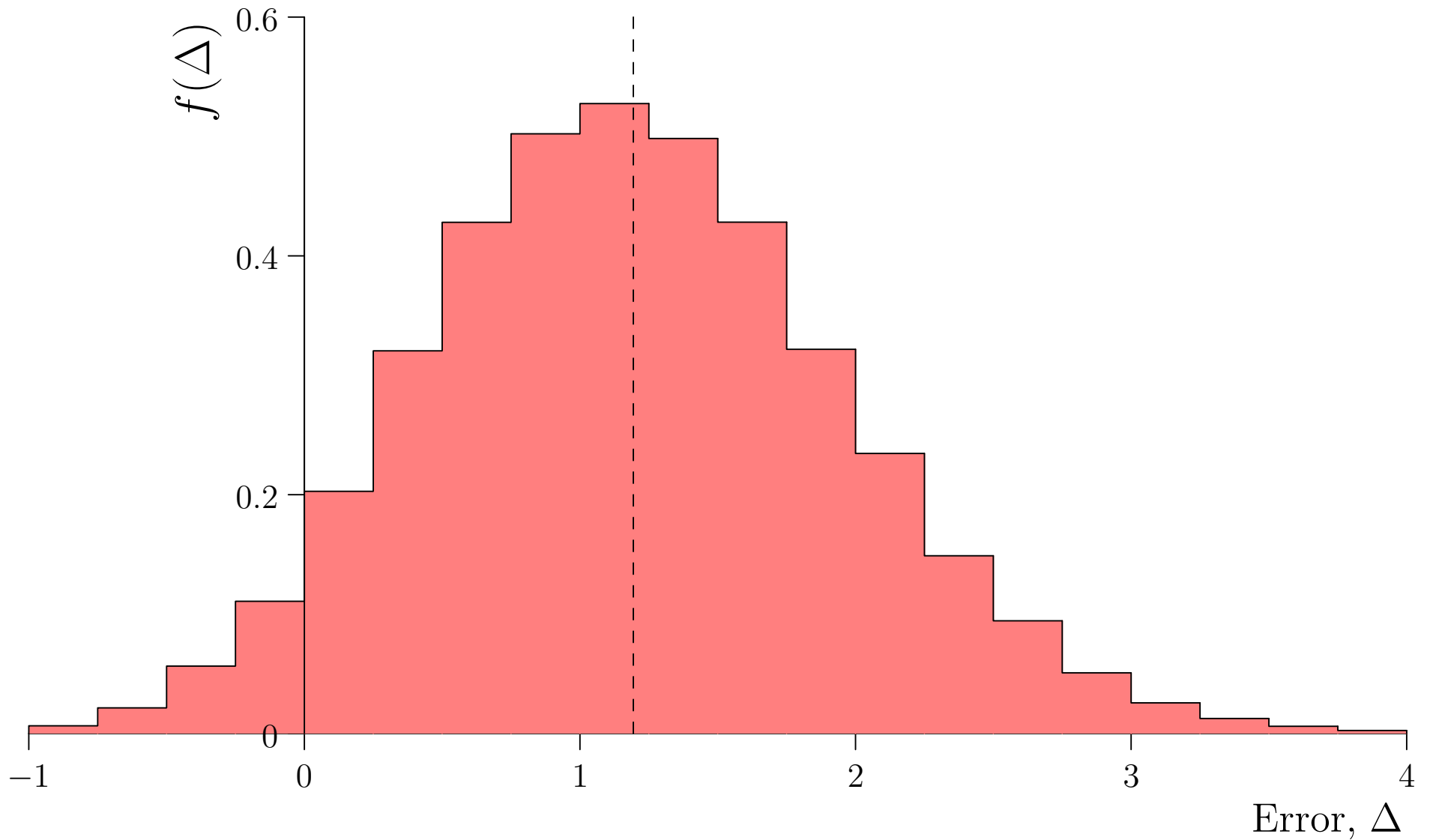
The Overfitting Game



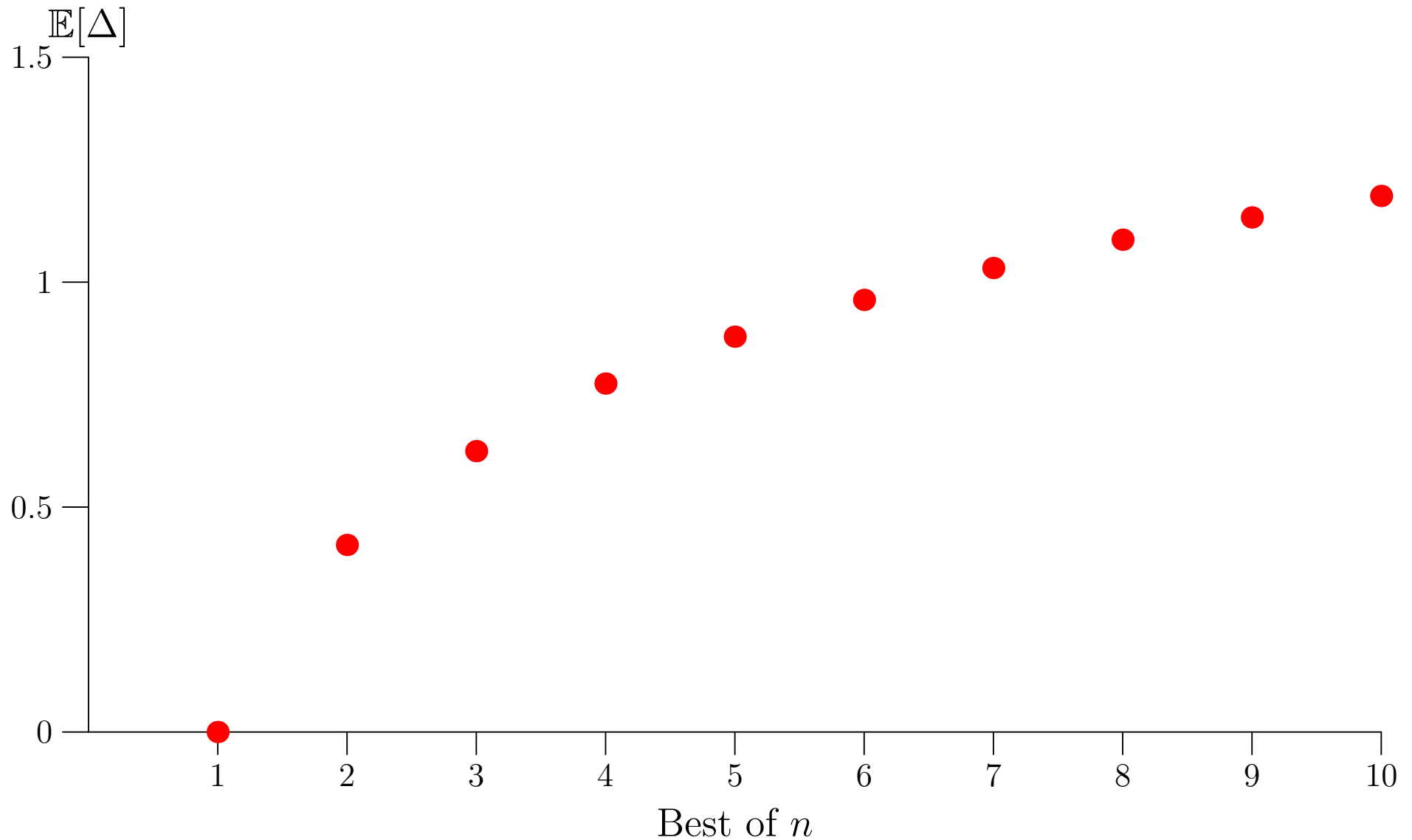
The Overfitting Game



The Overfitting Game



The Overfitting Game



Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test Set

Training Set

5-fold cross-validation

$$E_g = 5.1$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Training Set

Test Set

Training Set

5-fold cross-validation

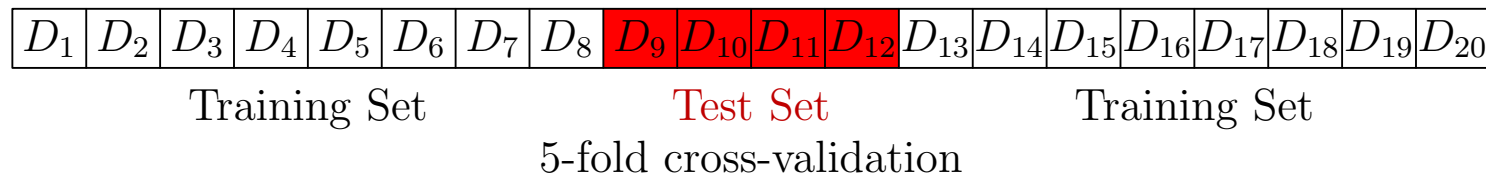
$E_g =$

3.7

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



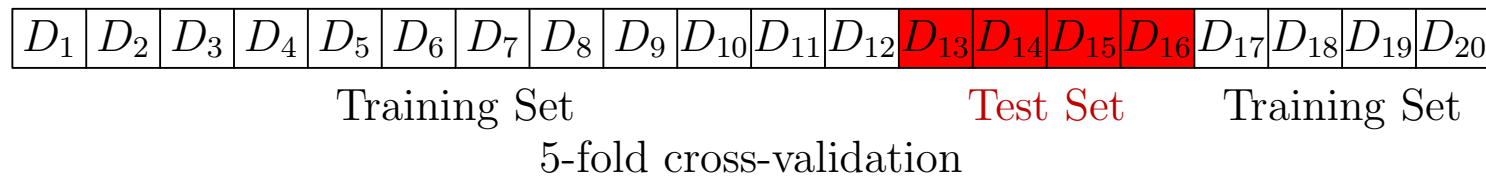
$E_g =$

4.6

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



$E_g =$

4.6

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Training Set

Test Set

5-fold cross-validation

$E_g =$

3.3

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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$$\langle E_g \rangle = \frac{5.1 + 3.7 + 4.6 + 4.6 + 3.3}{5} = 4.3$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test Set

Training Set

10-fold cross-validation

$$E_g = 5.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test Set

Training Set

10-fold cross-validation

$$E_g = 1.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Training Set **Test Set**

Training Set

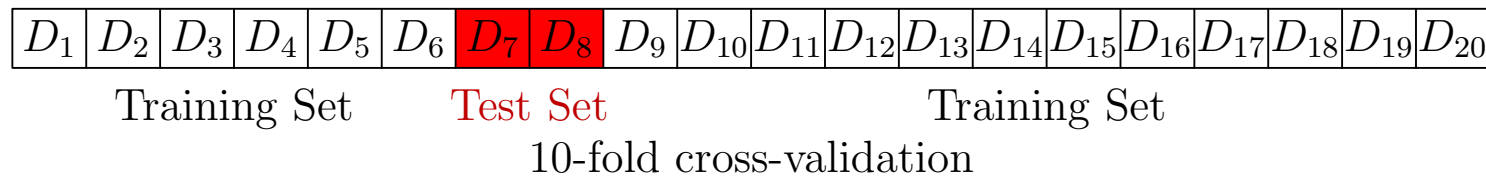
10-fold cross-validation

$$E_g = 4.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

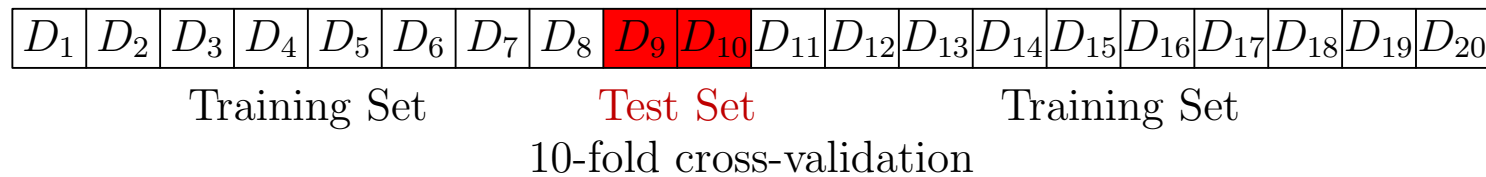


$$E_g = 3.6$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



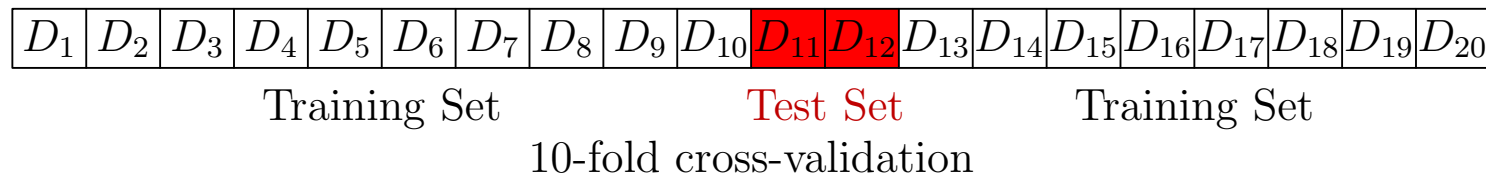
$E_g =$

7.4

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

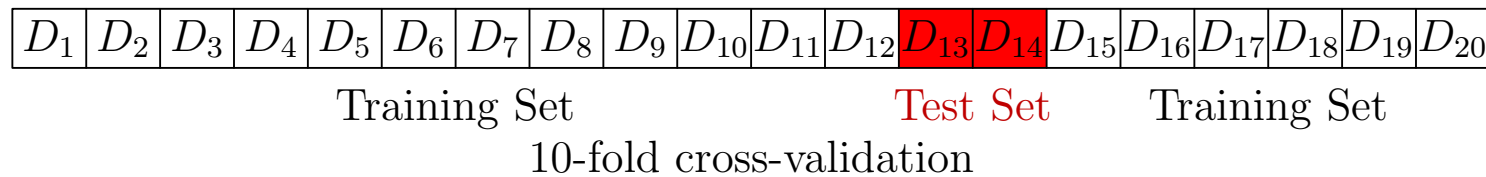


$$E_g = 0.99$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



$E_g =$

4.5

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Training Set

Test Set

Training Set

10-fold cross-validation

$E_g =$

5.4

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Training Set

Test Set

10-fold cross-validation

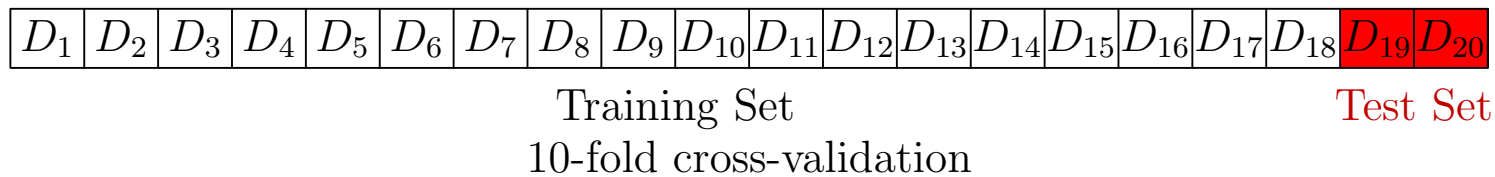
$E_g =$

6.2

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



$E_g =$

2.7

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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$$\langle E_g \rangle = \frac{5.8 + 1.8 + 4.8 + 3.6 + 7.4 + 0.99 + 4.5 + 5.4 + 6.2 + 2.7}{10} = 4.3$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test

Leave-one-out cross-validation

$$E_g = 5$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test

Leave-one-out cross-validation

$$E_g = 3.7$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test

Leave-one-out cross-validation

$$E_g = 5$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test

Leave-one-out cross-validation

$$E_g = 2.2$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test

Leave-one-out cross-validation

$$E_g = 3.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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$$\langle E_g \rangle = 3.9$$

Cross Validation

- If you want to use more data for training then you can use cross validation
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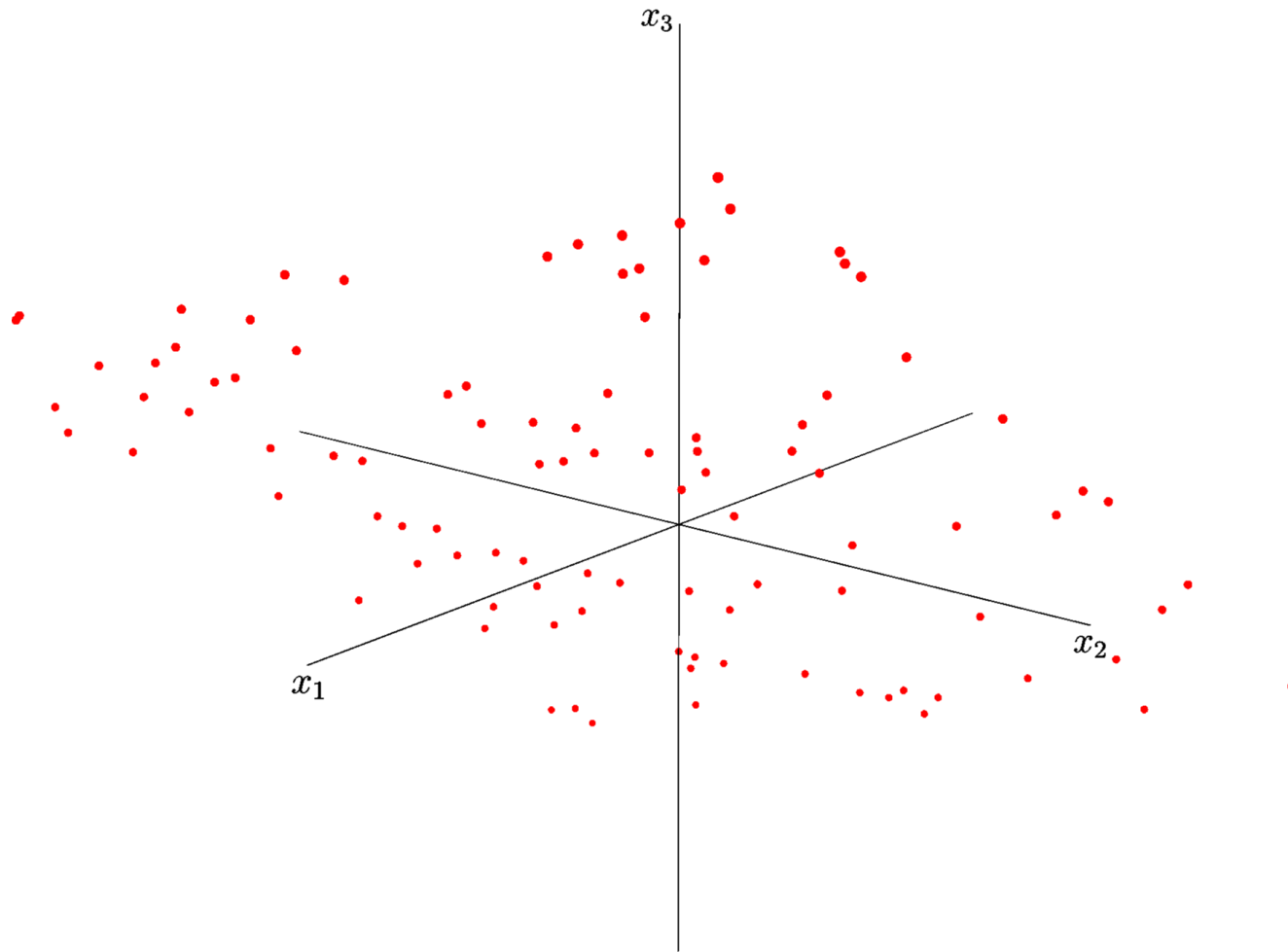
D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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$$\langle E_g \rangle = 3.9$$

- Leave-one-out cross-validation is extreme case

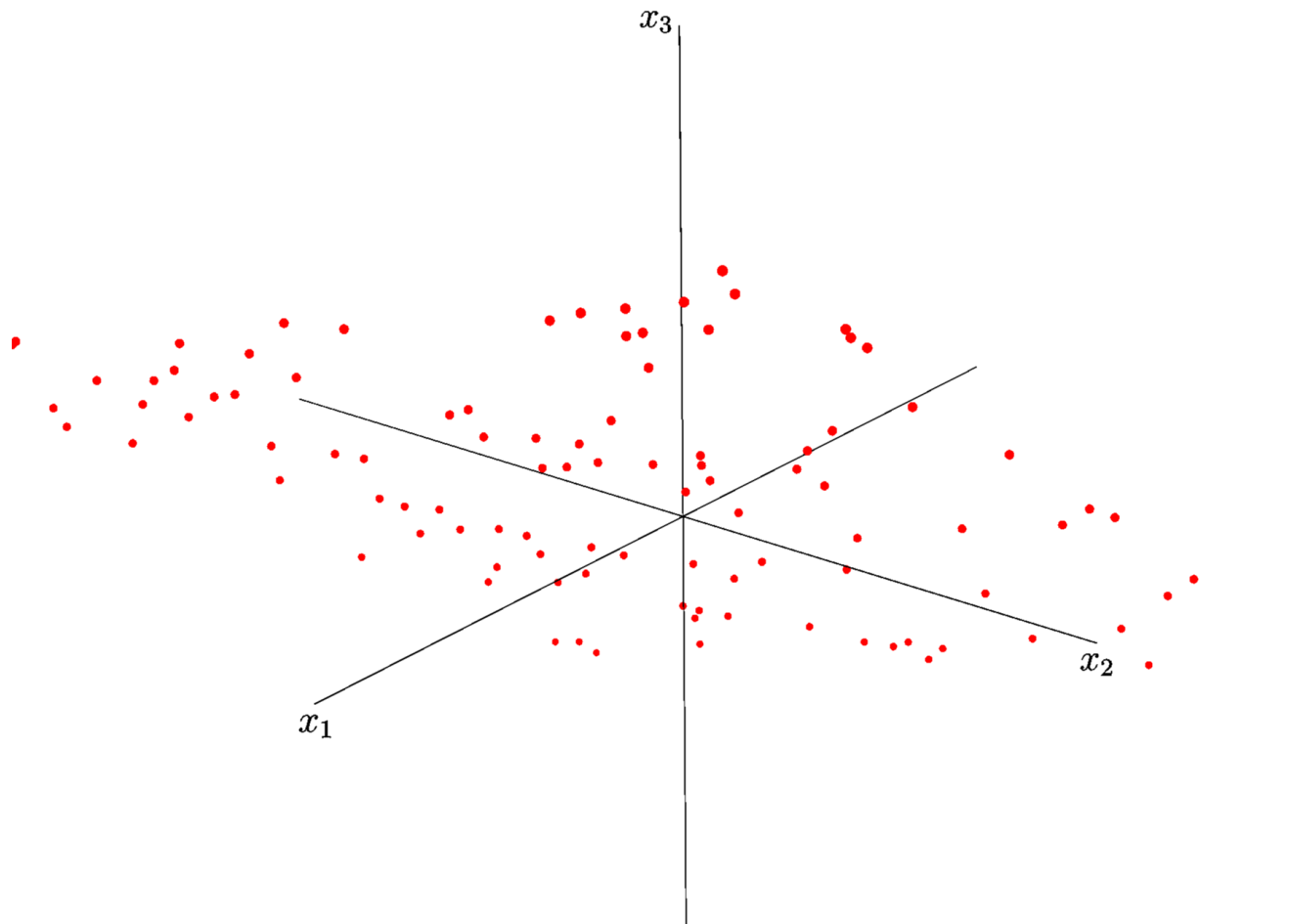
Hidden Structure

Can't spot low dimensional data by looking at numbers



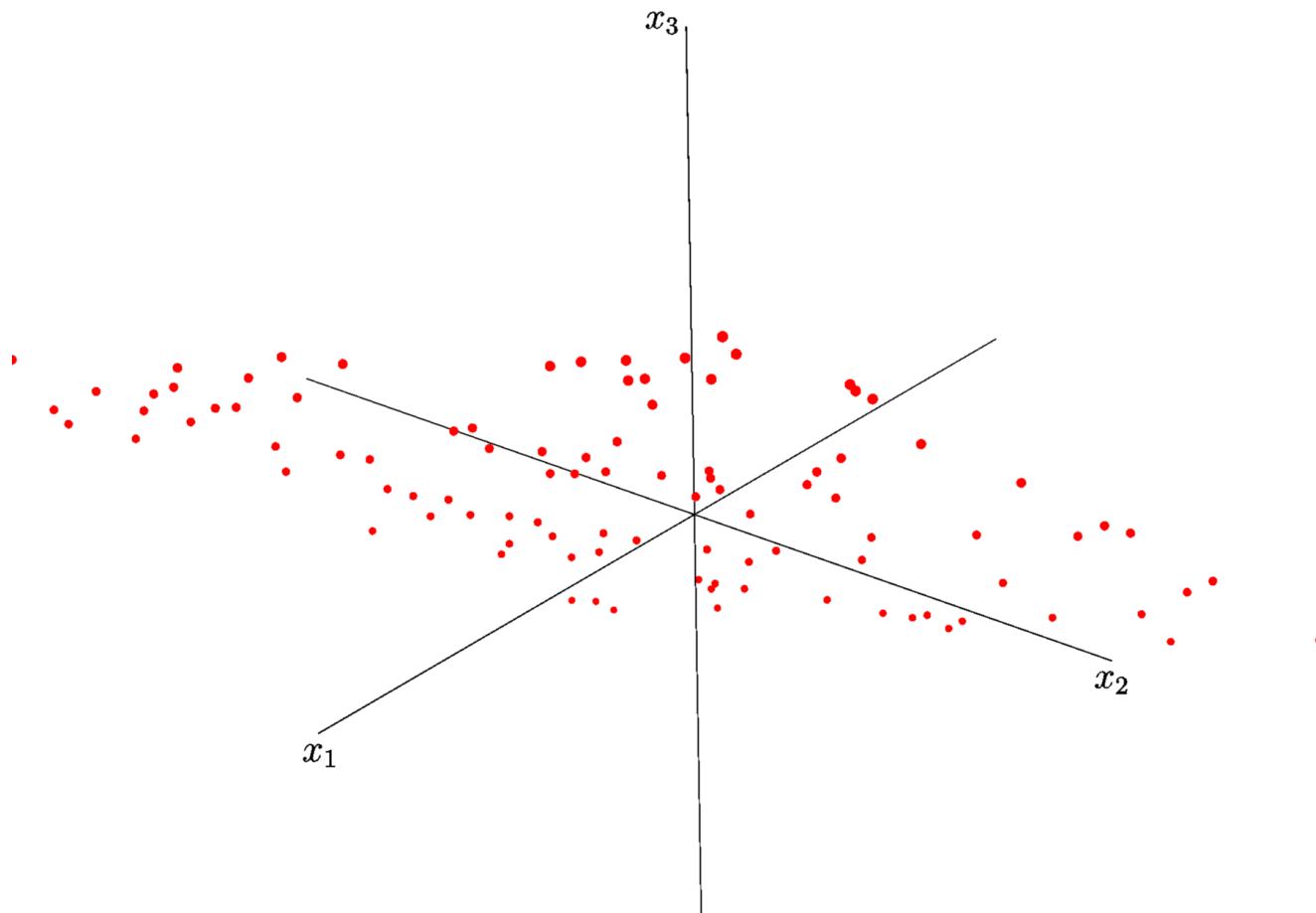
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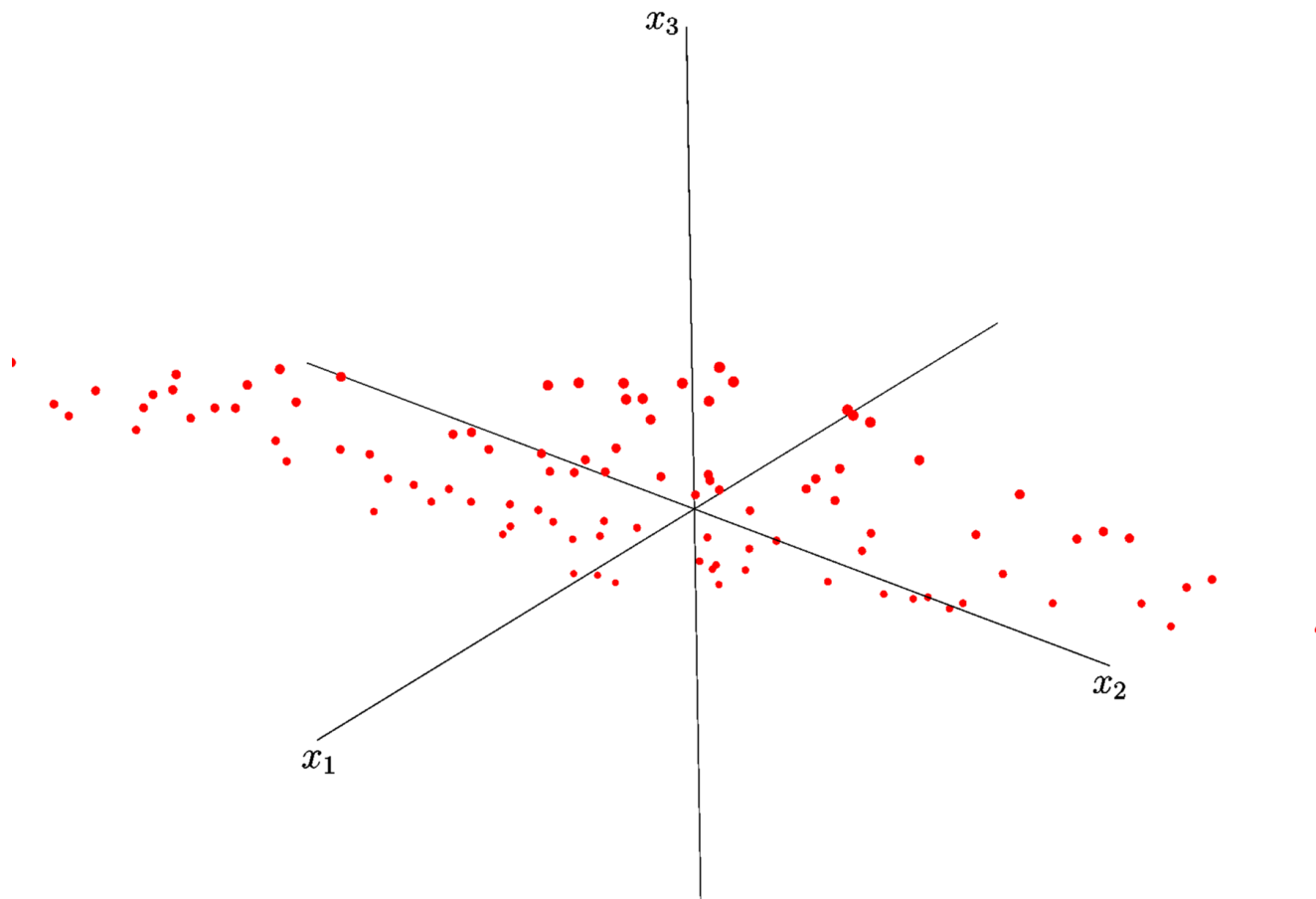
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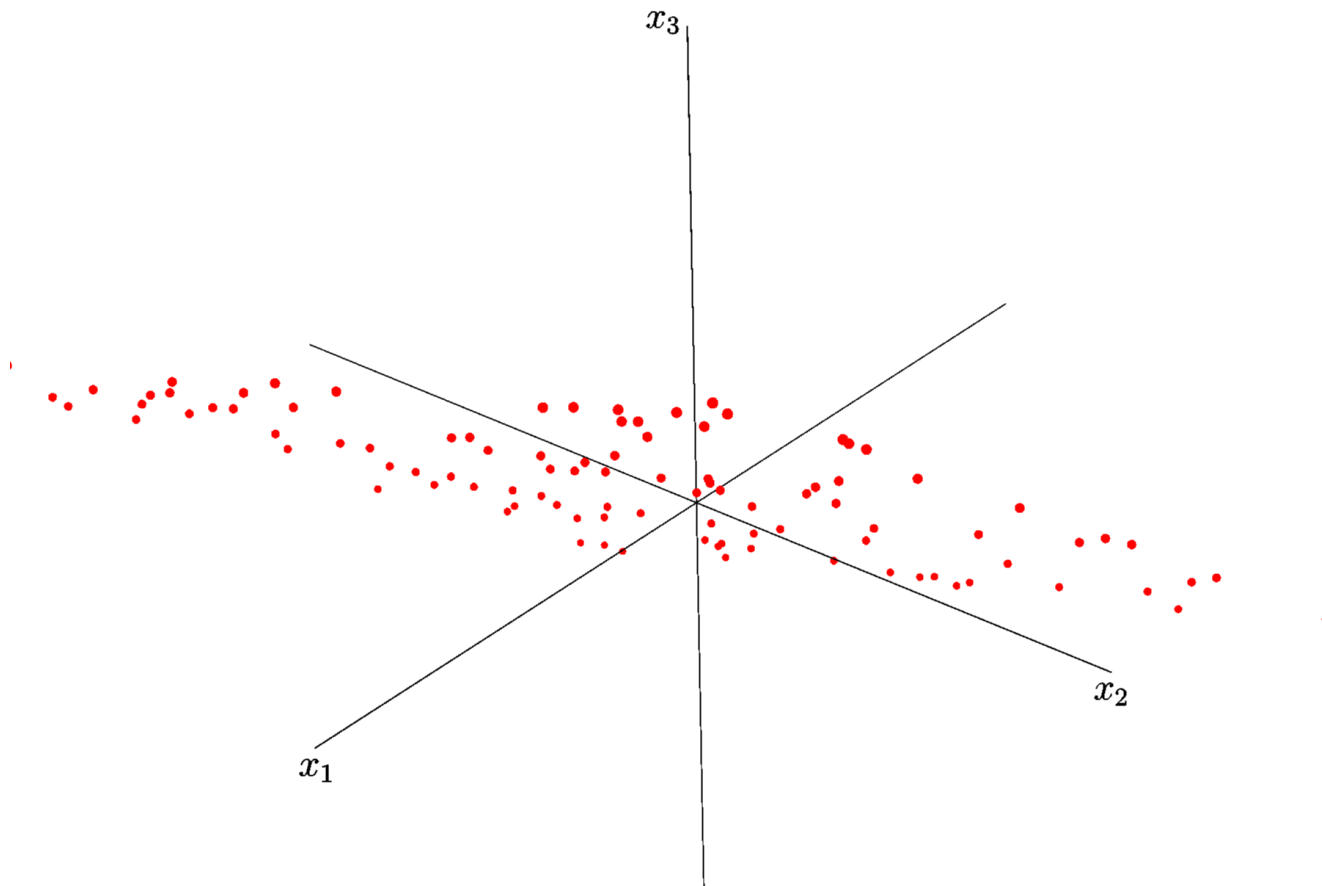
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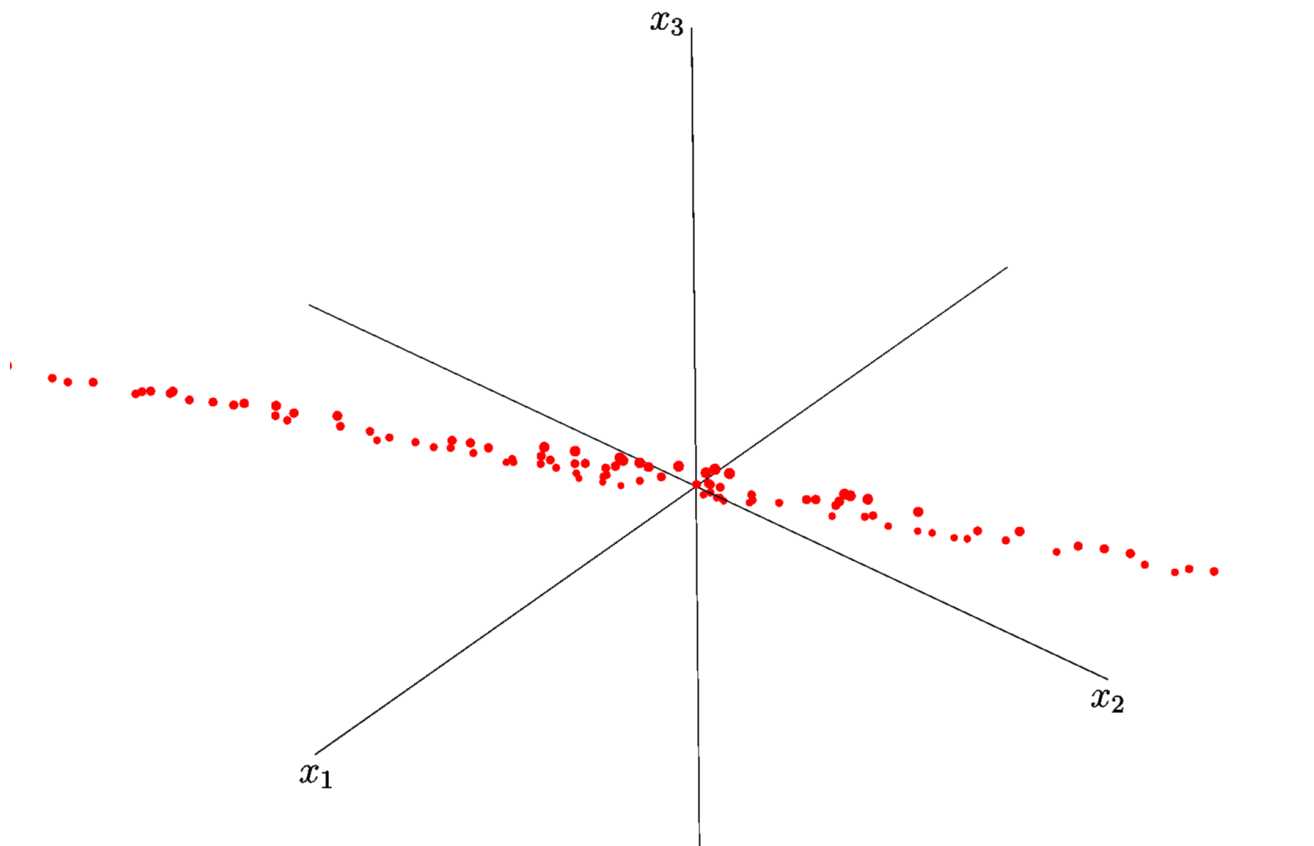
Hidden Structure

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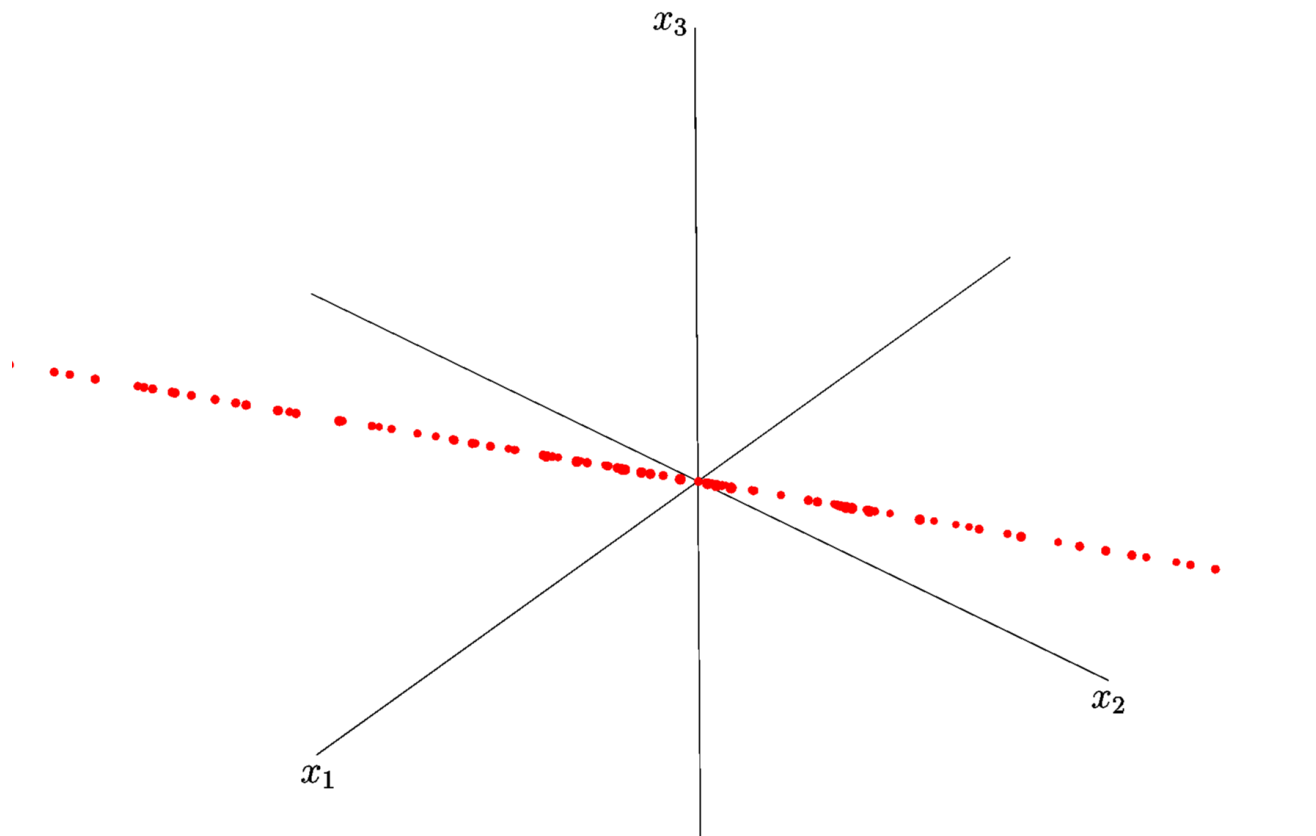
Hidden Structure

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Hidden Structure

Can't spot low dimensional data by looking at numbers



Dimensionality Reduction

- We can sometimes simplify our machines by using less features
- We can project our data onto a lower dimensional sub-space (e.g. one with the maximum variation in the data: PCA)
- We can use clustering to find exemplars and recode our data in terms of distances from the exemplars (radial basis functions)
- Whether this helps depends on whether the information we discard is pertinent to the task we are trying to perform

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Feature Selection

- Spurious features will allow us to find spurious rules (**over-fitting**)
- We can try different combinations of features to find the best set, although it rapidly becomes intractable to do this in all ways
- We can use various heuristics to decide which features to keep, but no heuristic is fail-safe method to find the best set of features
- Feature selection however can be powerful, often we can get very good results by keeping only a few variables
- As well as possibly improving generalisation we also get a more **interpretable** rule

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Normalising Features

- Measuring a feature in millimeters or kilometers is going to make a lot of difference to the size of that feature
- Many learning algorithms are sensitive to the size of a feature (larger features are more important)
- If we don't know how important different features are then it makes sense to normalise our features

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Normalising Features

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- If we don't know how important different features are then it makes sense to normalise our features, E.g.

$$x_i^\alpha \leftarrow \frac{x_i^\alpha - \hat{\mu}_i}{\hat{\sigma}_i}, \quad \hat{\mu} = \frac{1}{m} \sum_{\beta=1}^m x_i^\beta, \quad \hat{\sigma}_i^2 = \frac{1}{m-1} \sum_{\beta=1}^m (x_i^\beta - \hat{\mu}_i)^2$$

Outline

1. Over-fitting?
2. Controlling Complexity
3. Hidden structure
4. **Regularisation**



Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2$$

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$$

varies rapidly as we change x_i

Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \|\mathbf{w}\|^2$$

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$$

varies rapidly as we change x_i

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(Good to normalise features)

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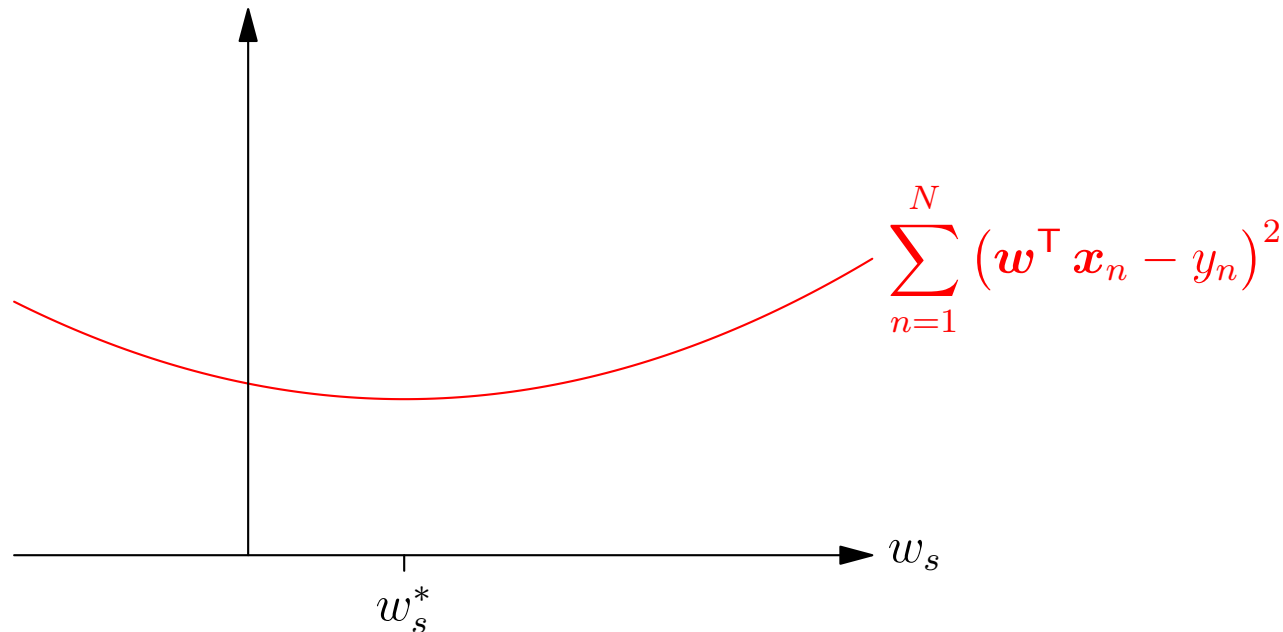
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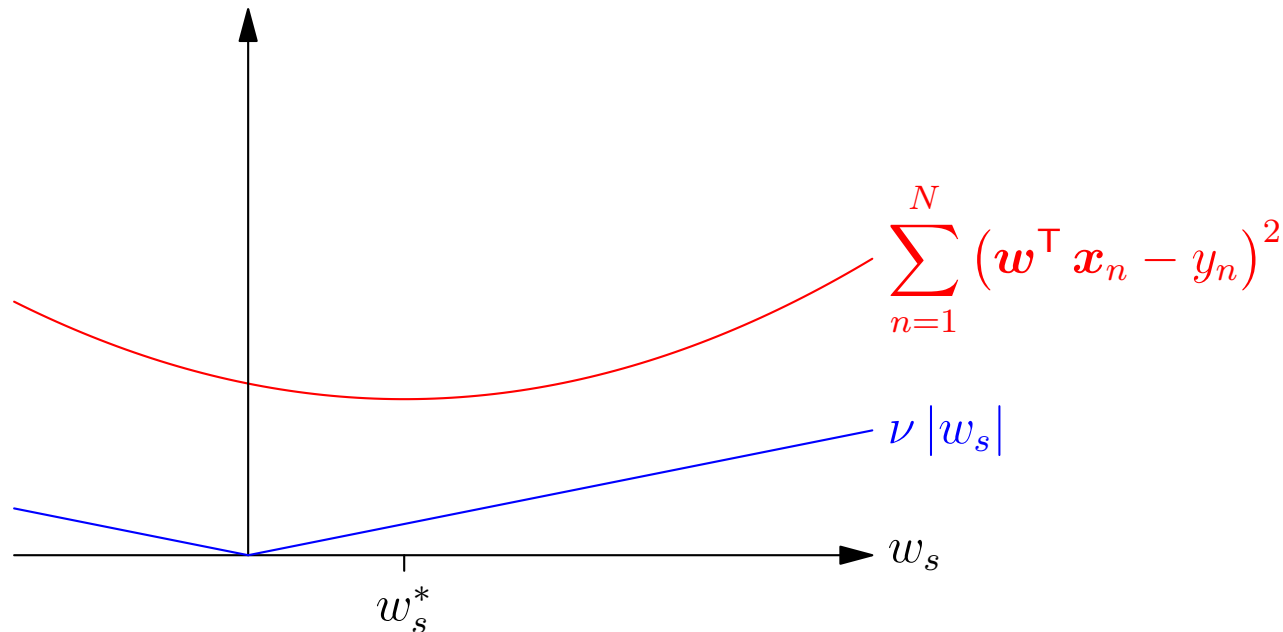


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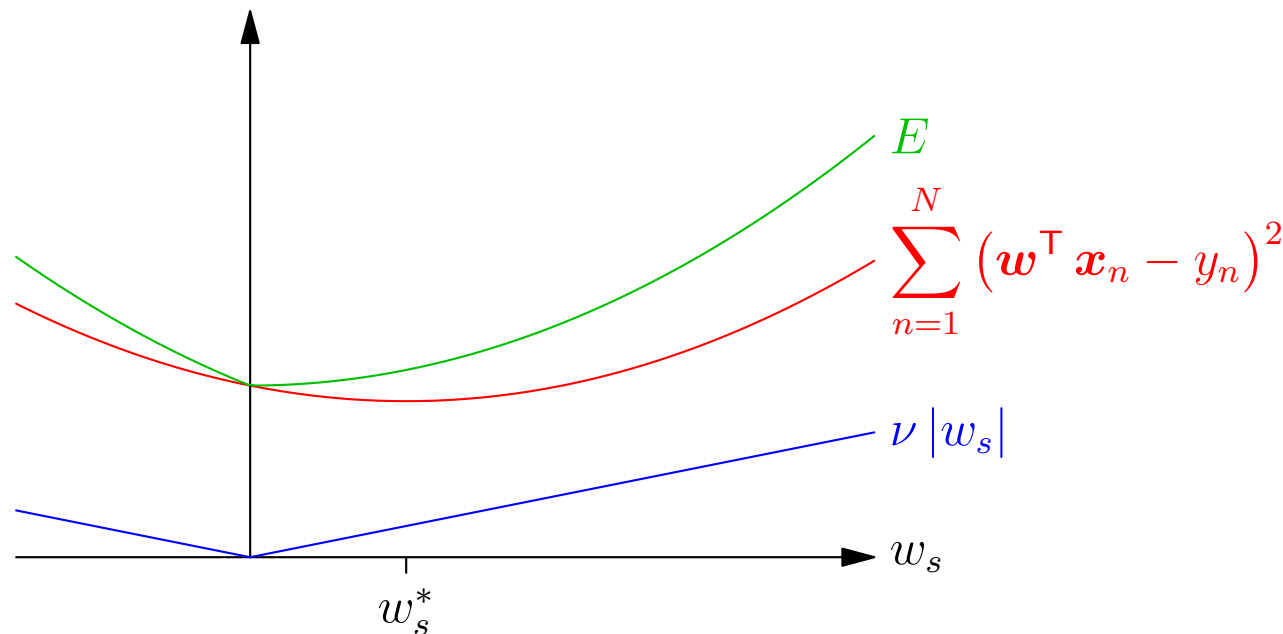


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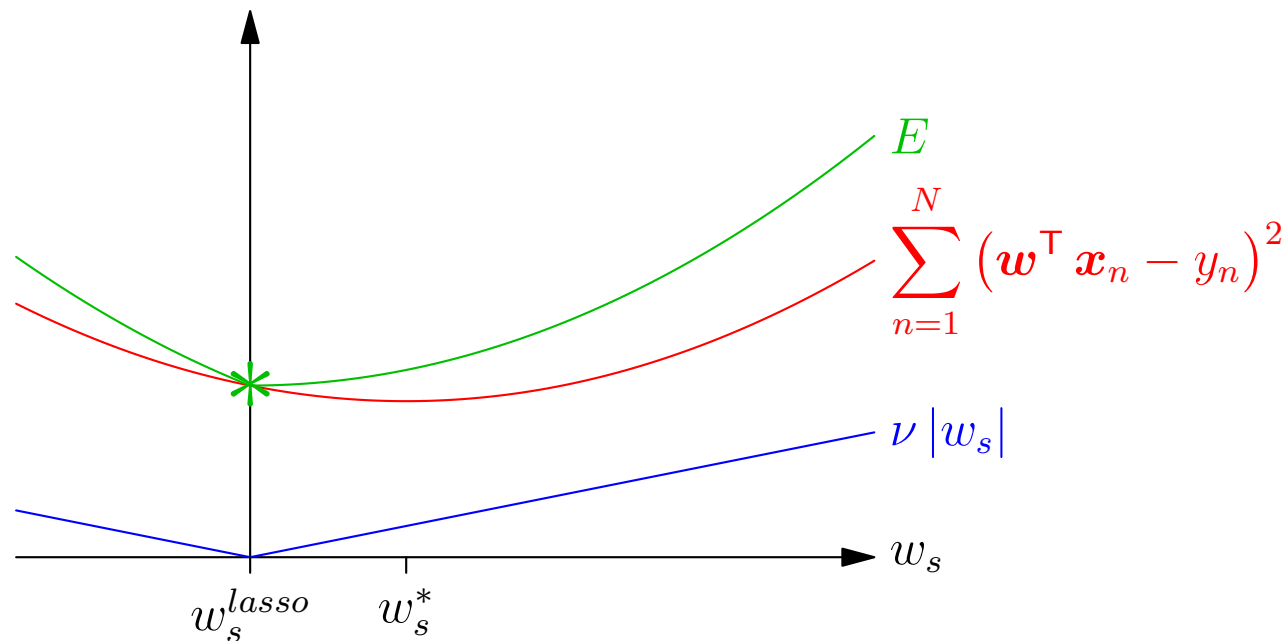


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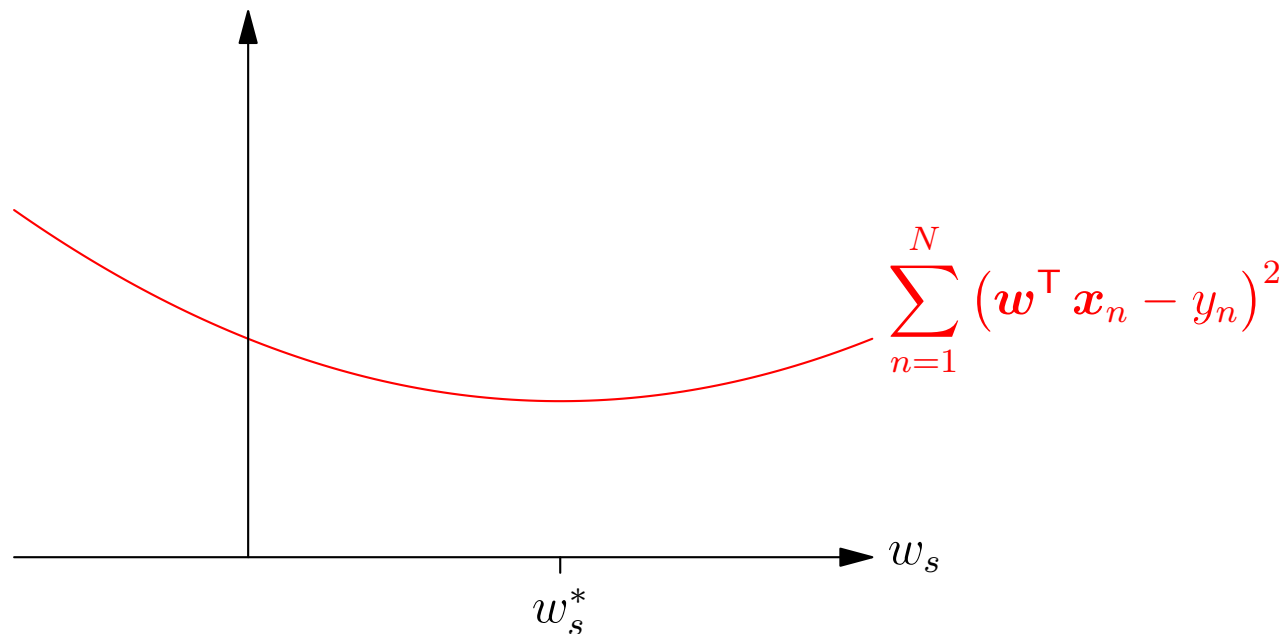


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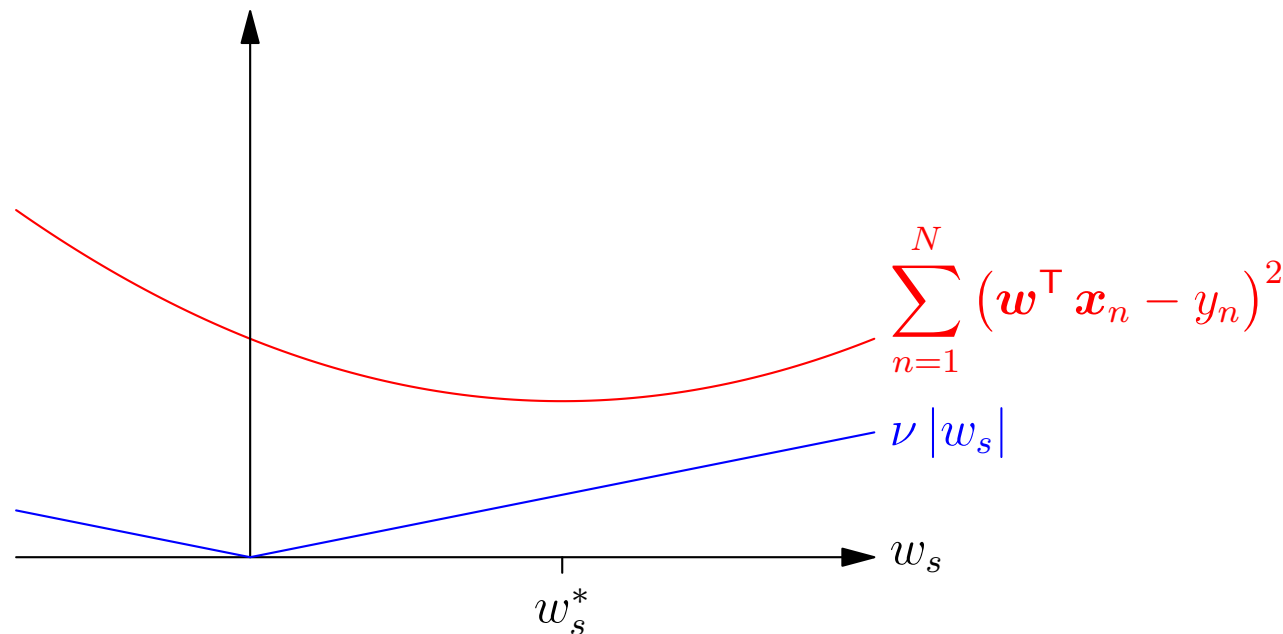


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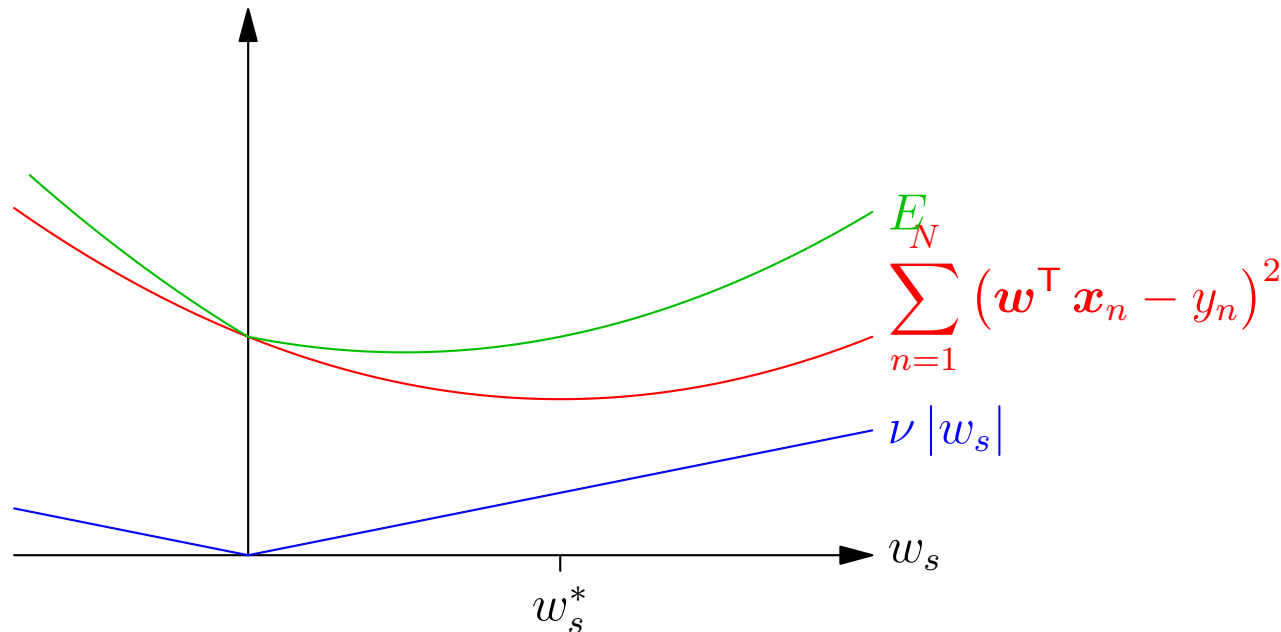


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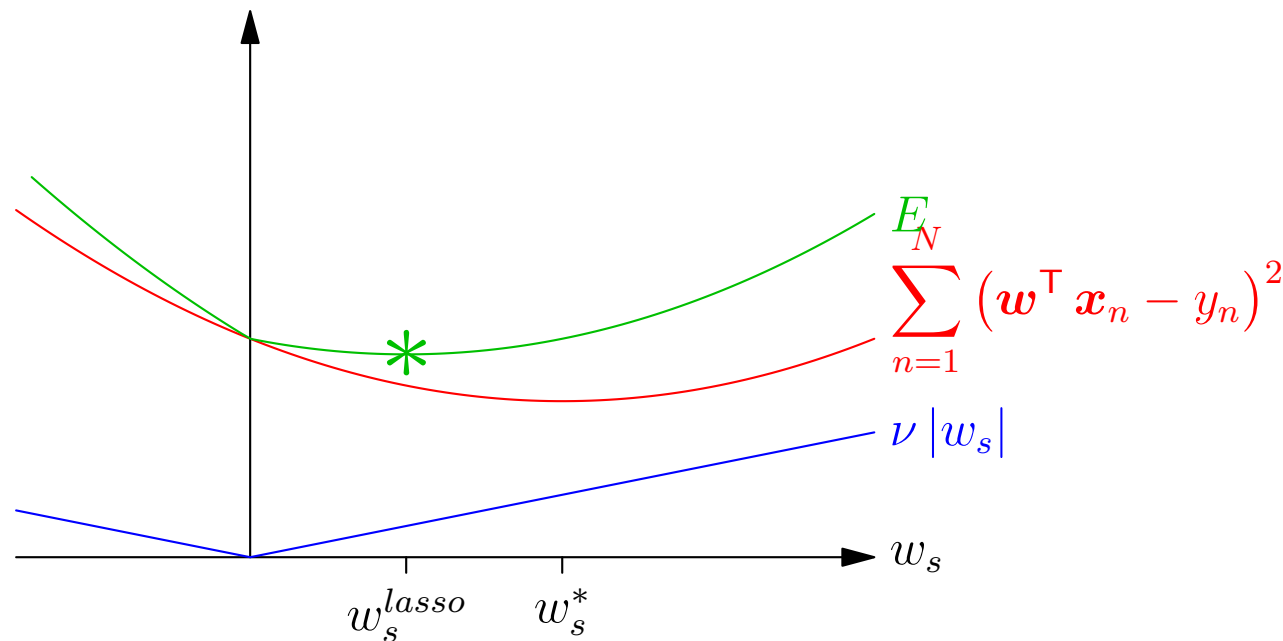


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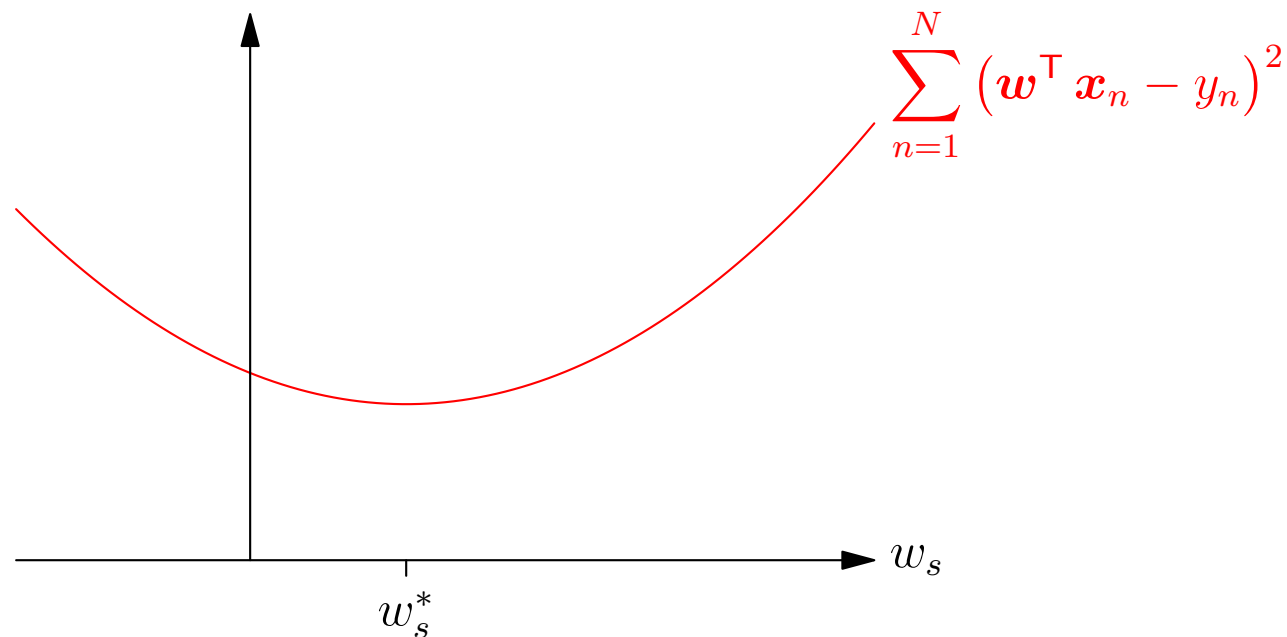


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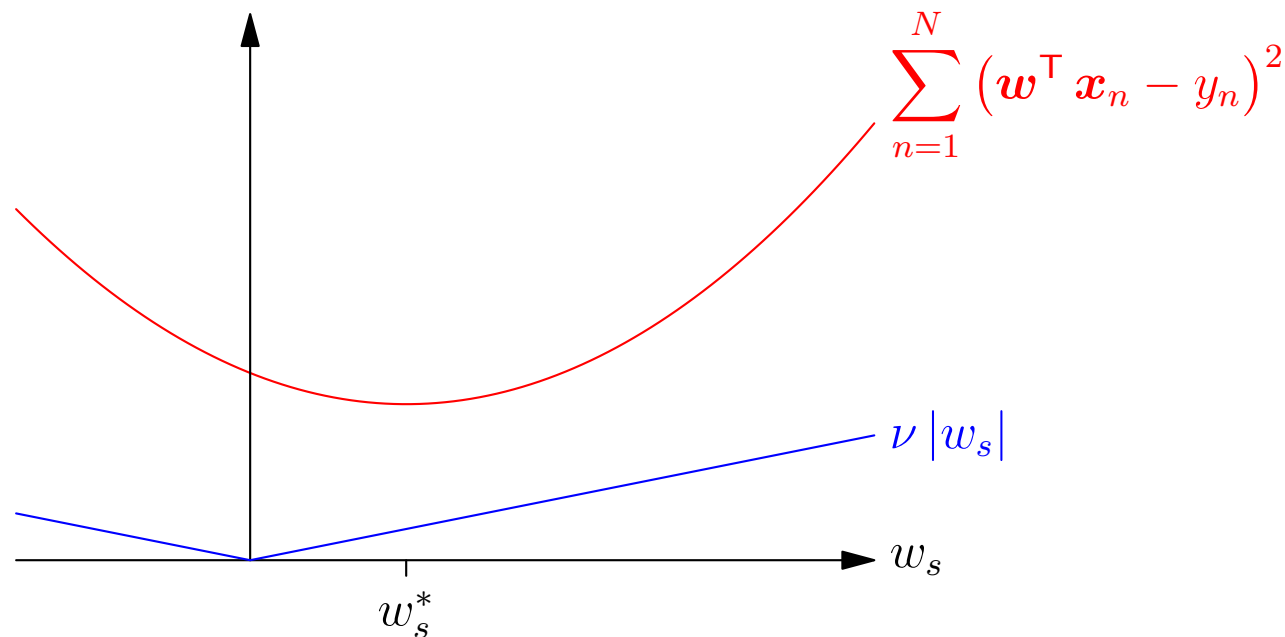


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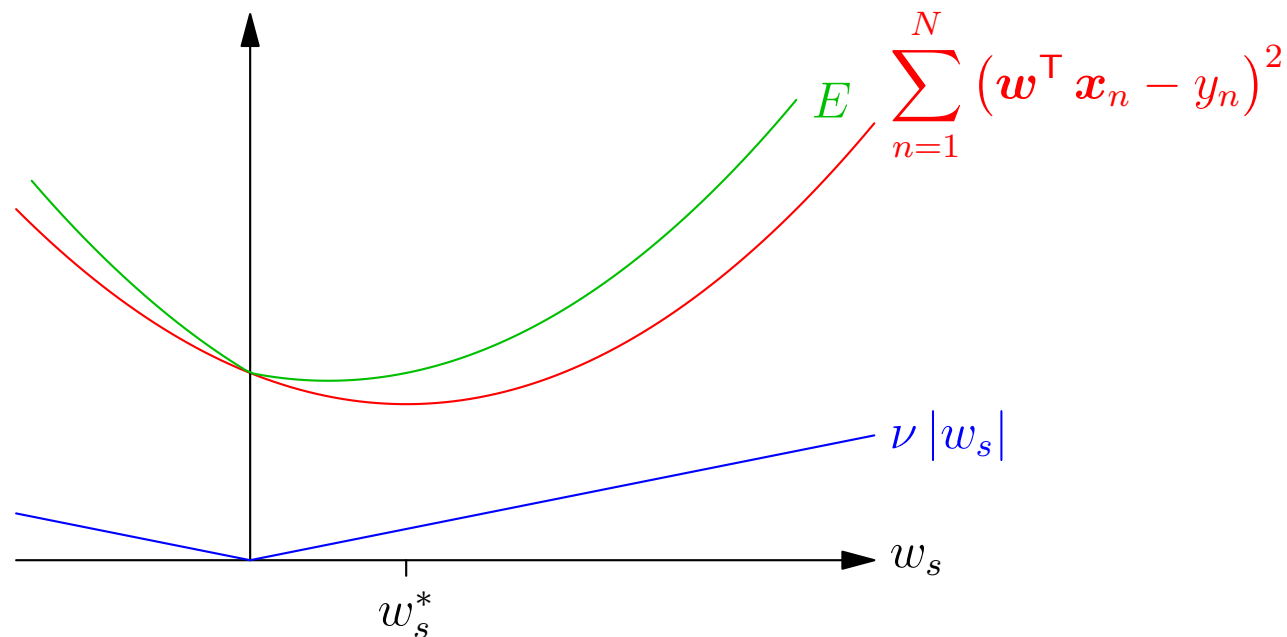


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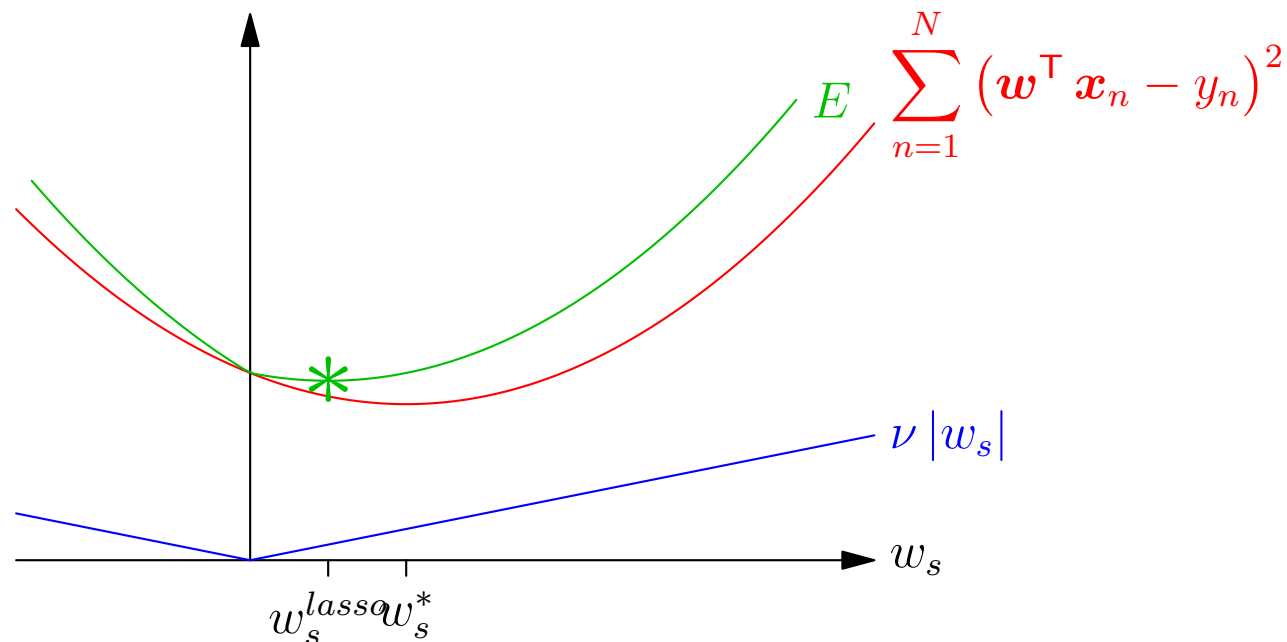


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- In the last two examples we added an explicit regularisation term that made the function we learnt simpler
- Some learning machines do this less explicitly
- Some deep learning architectures do subtle averaging
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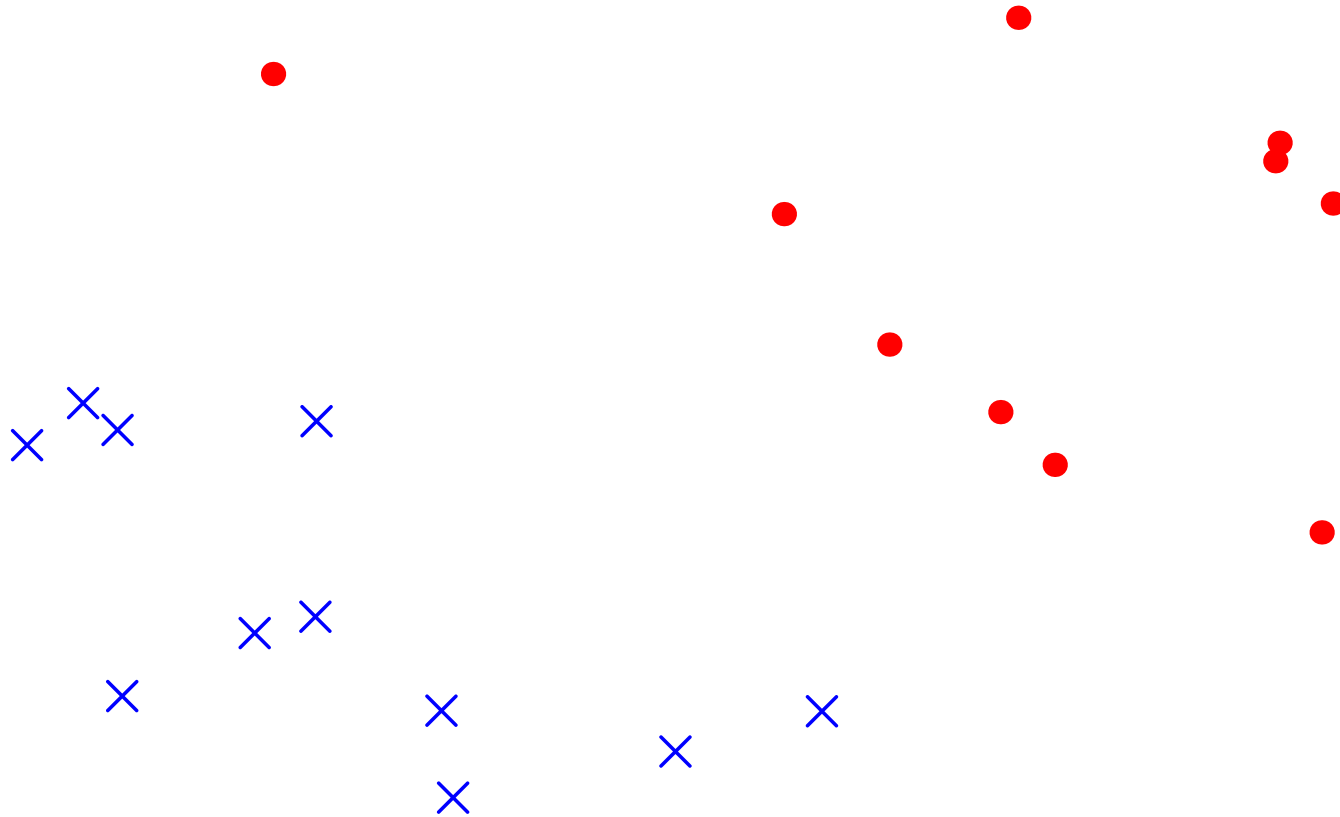
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- Perceptrons have many options to separate data

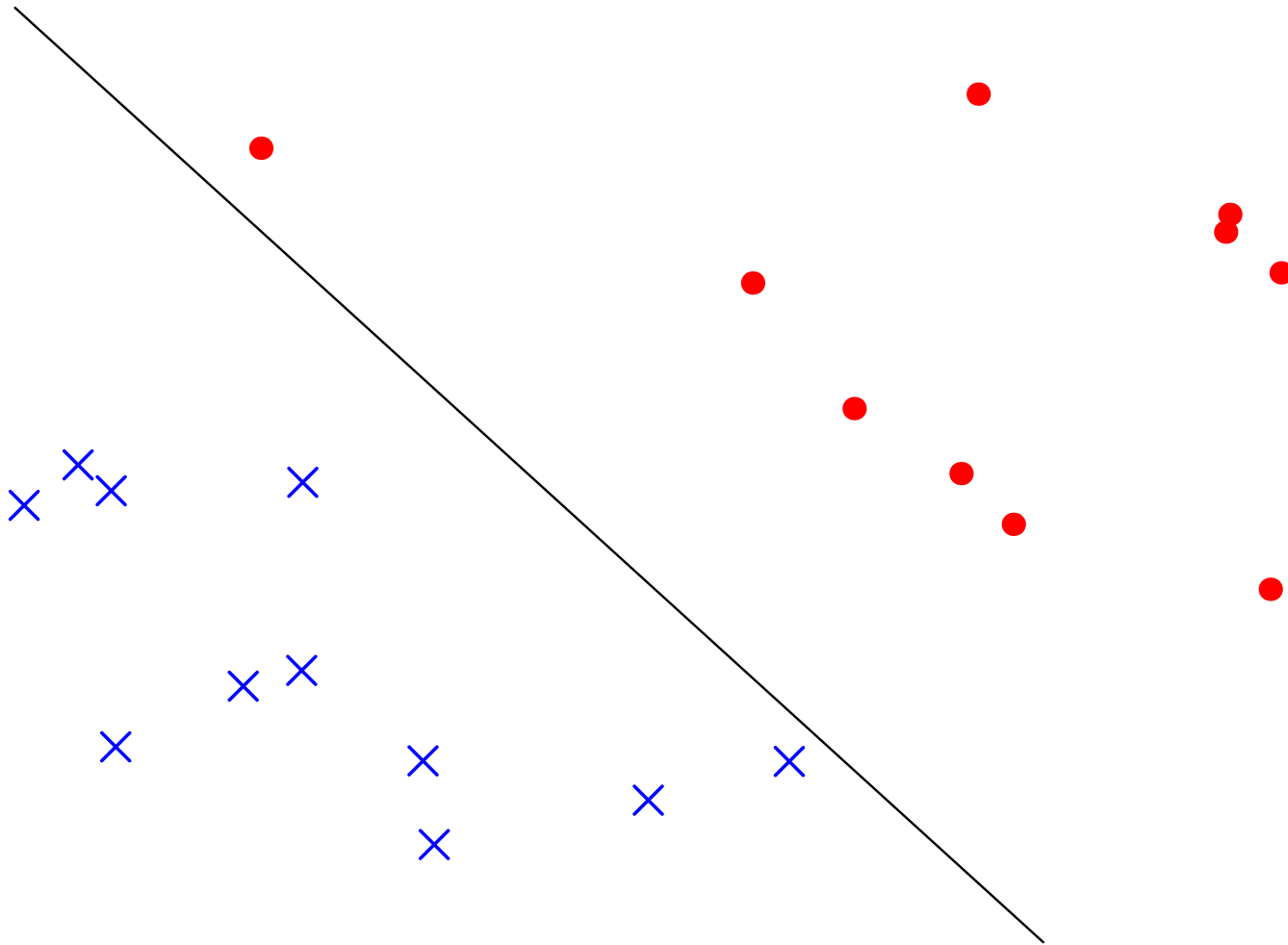
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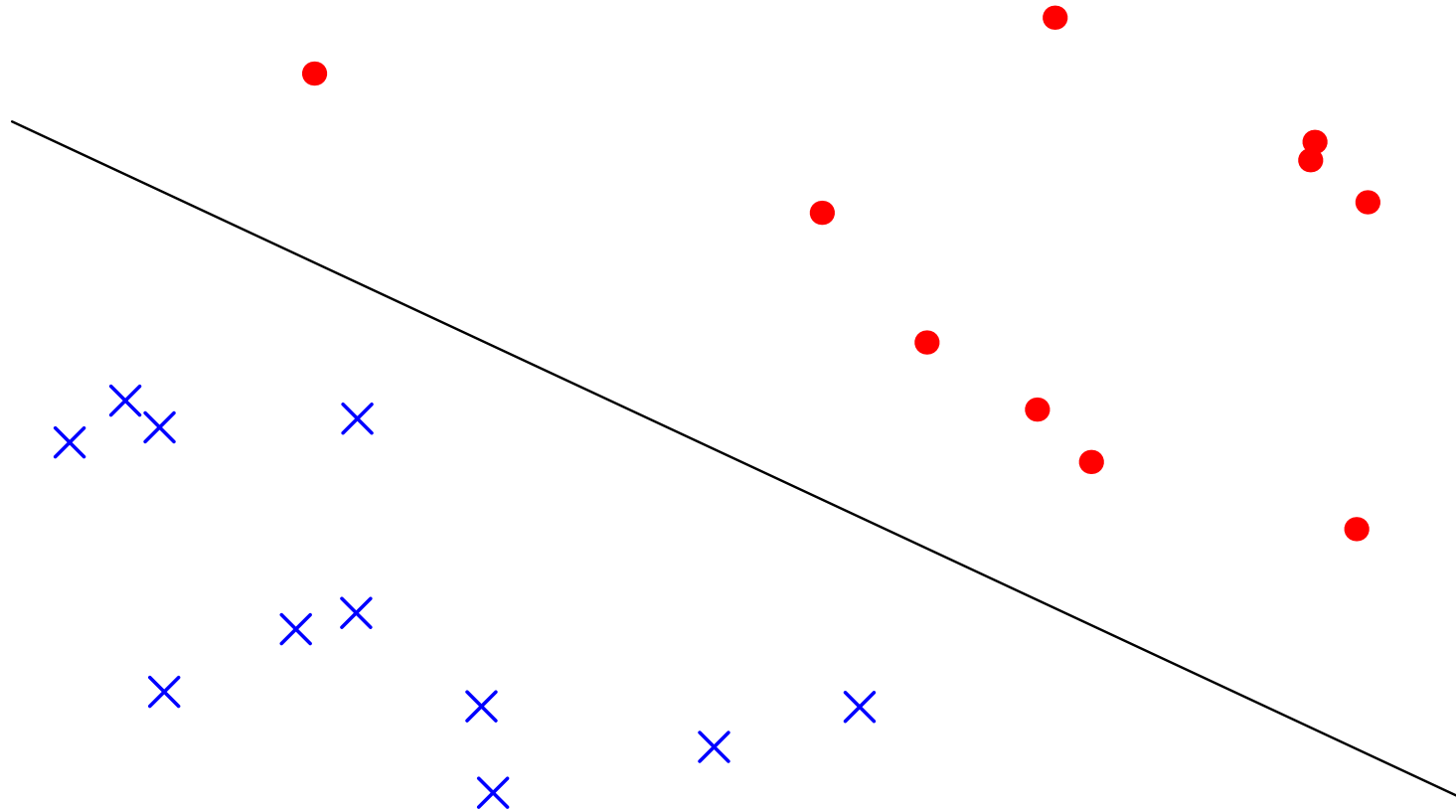
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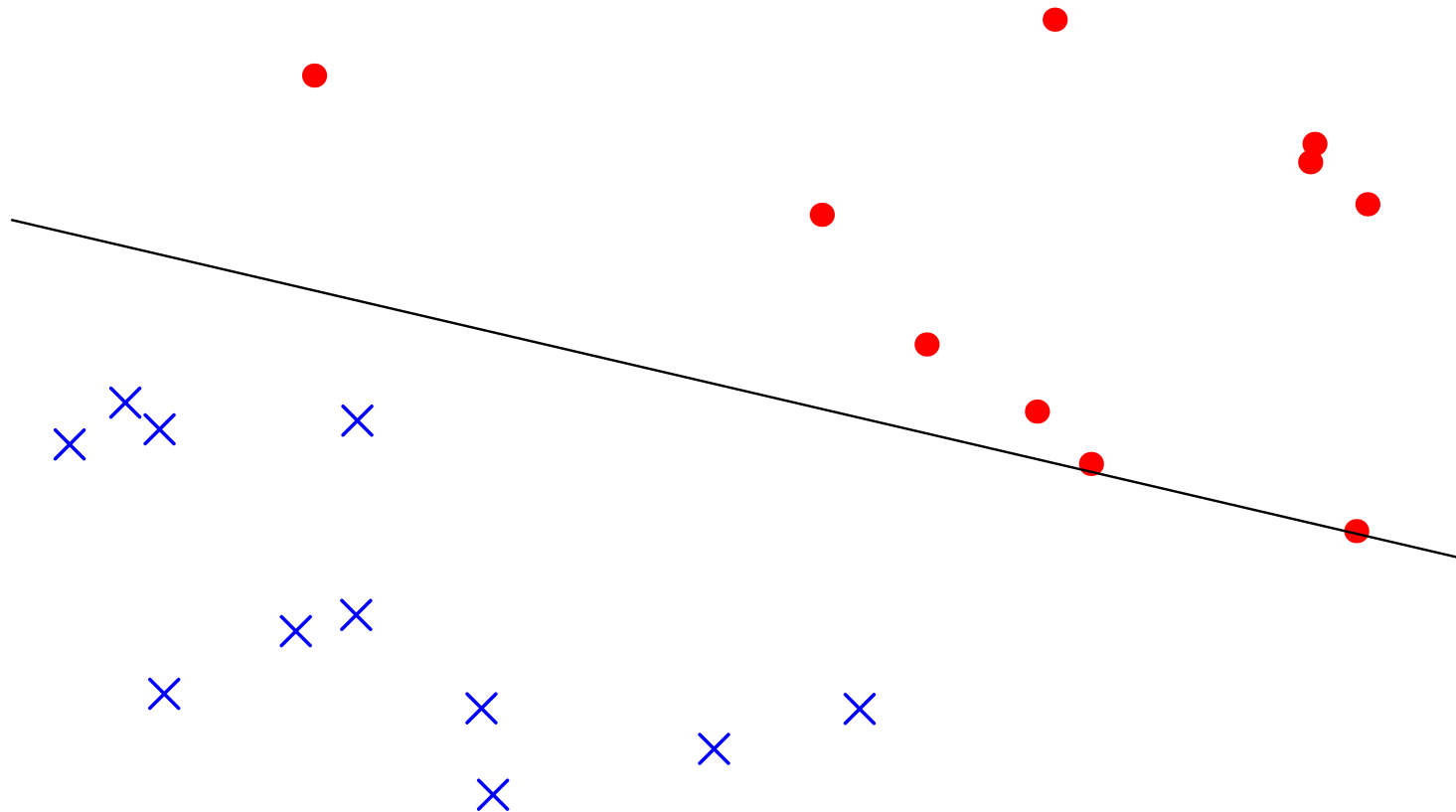
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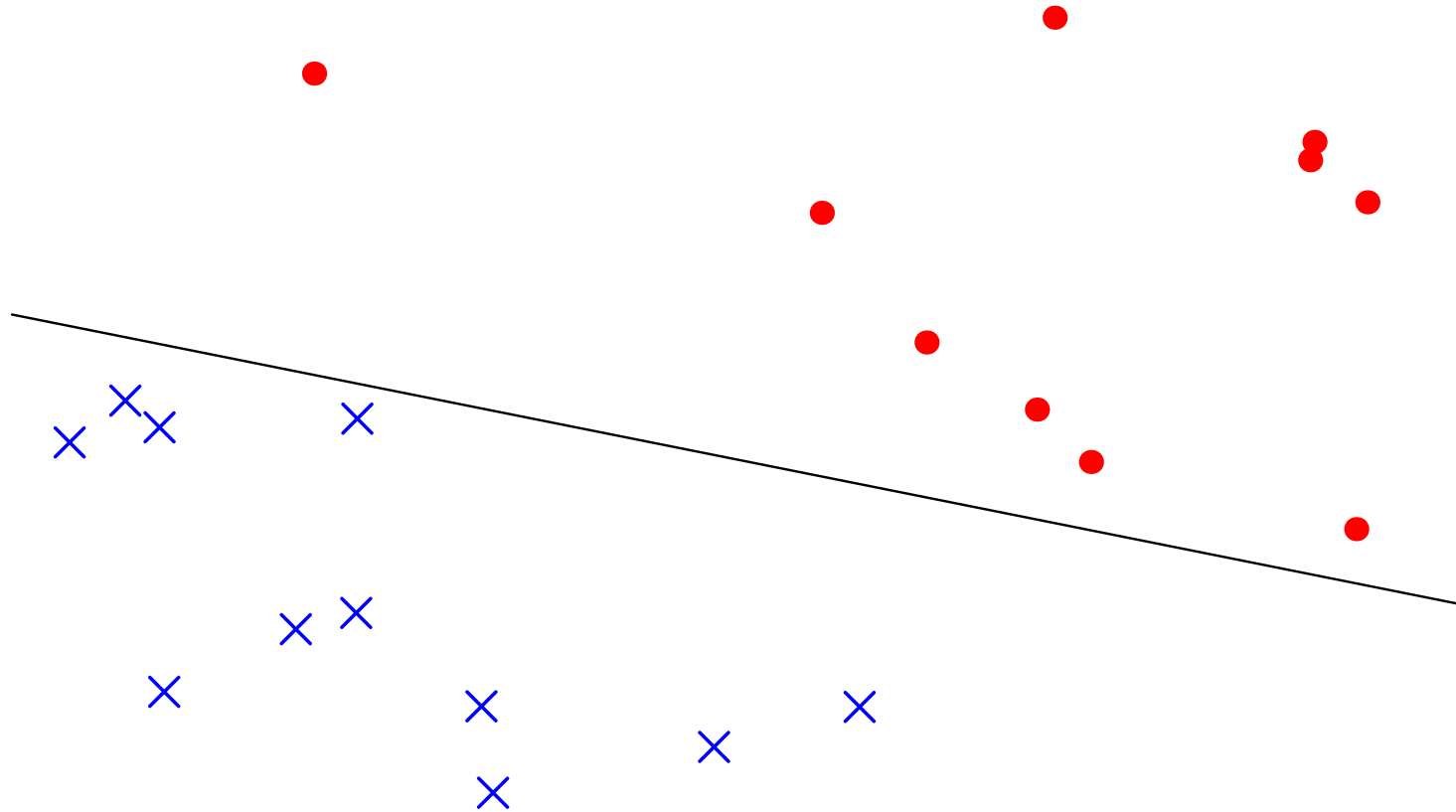
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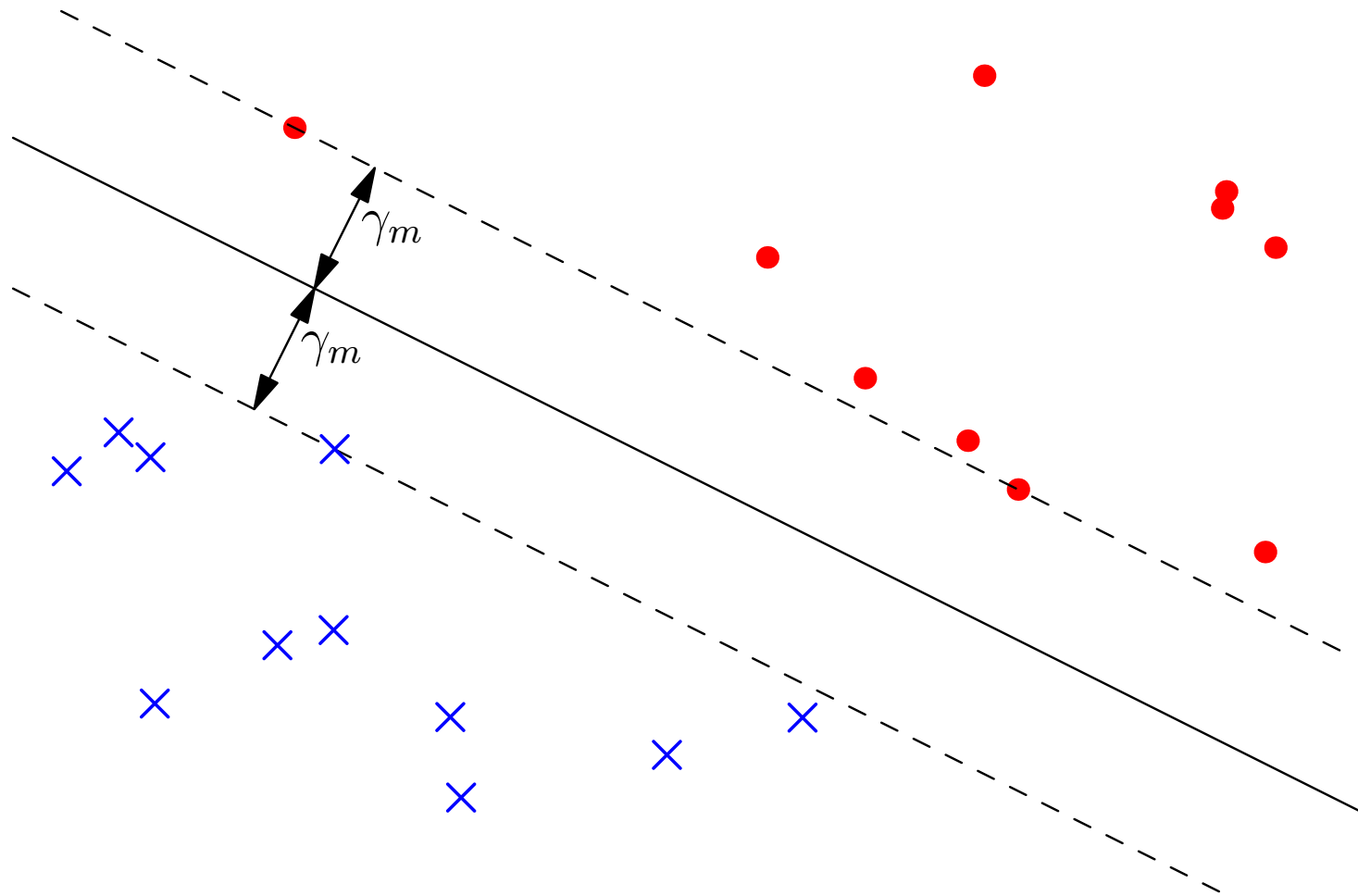
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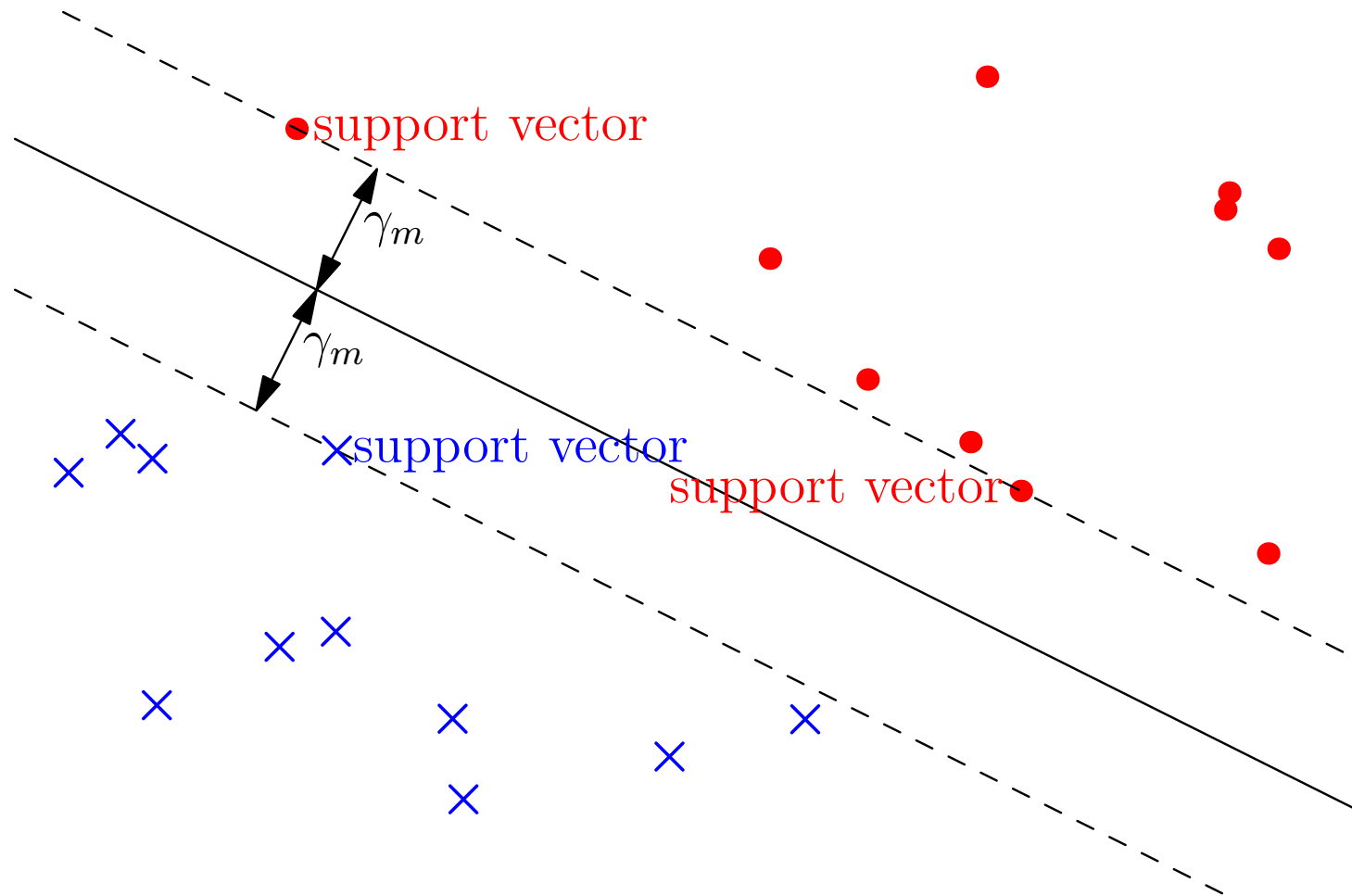
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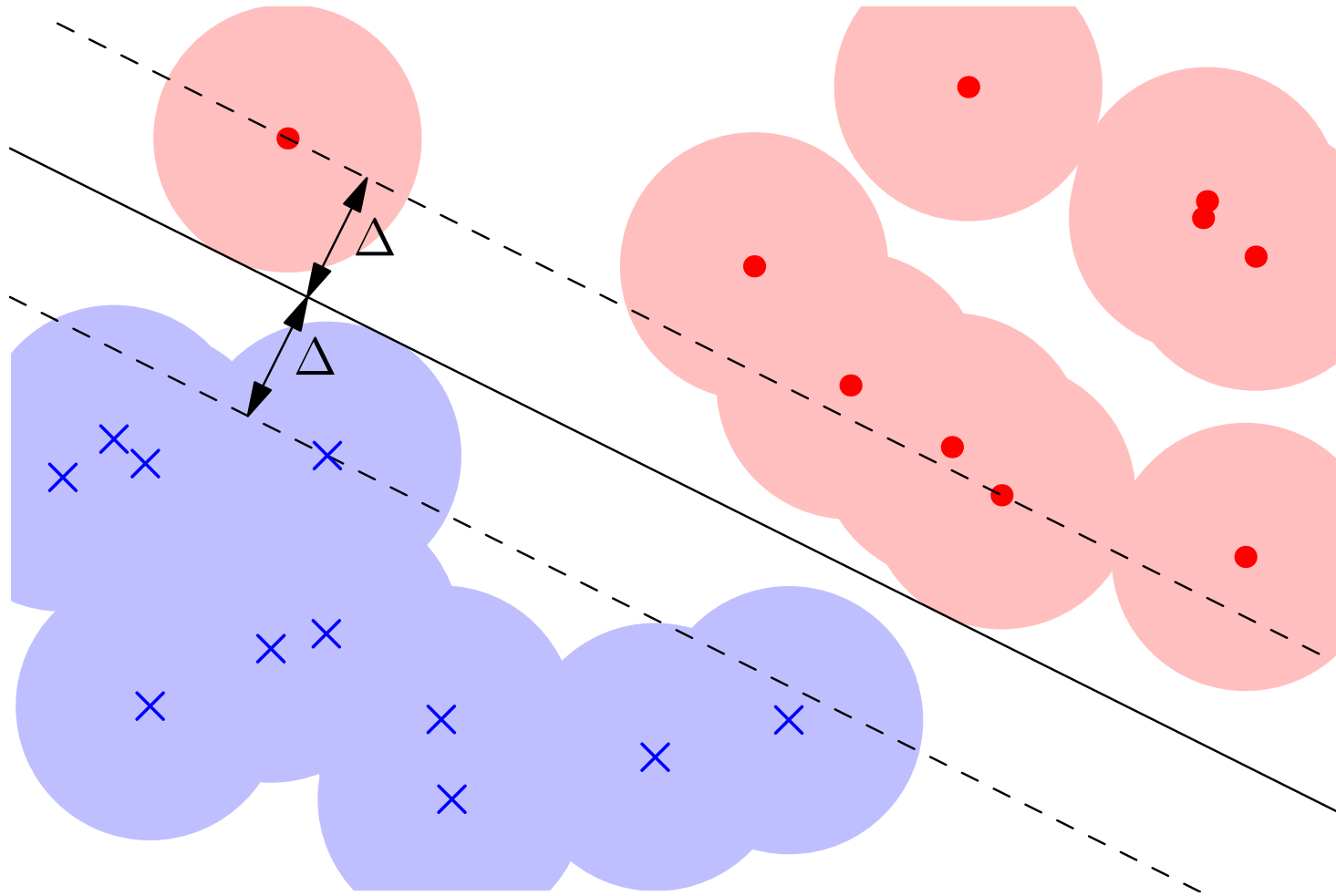
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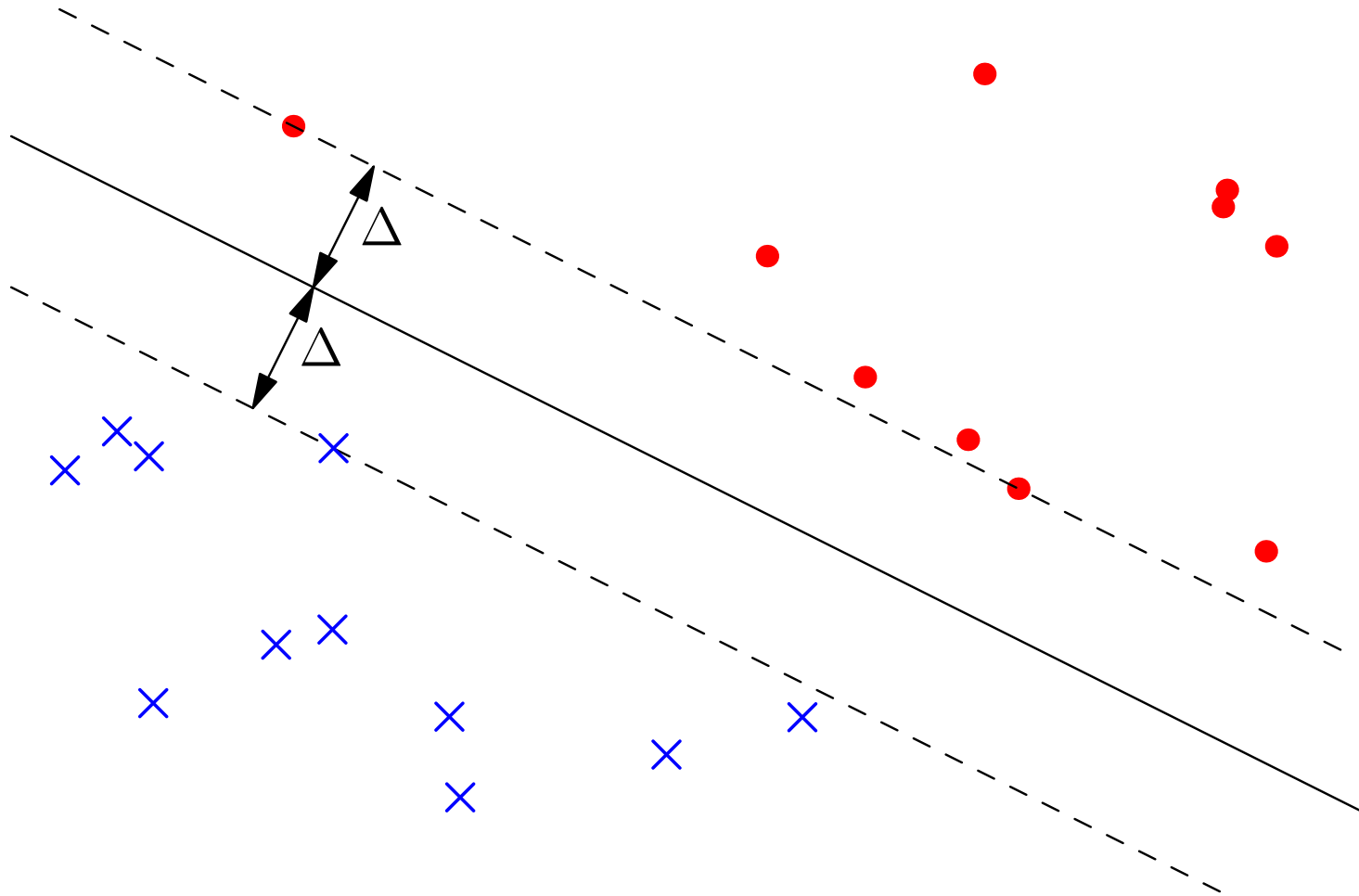
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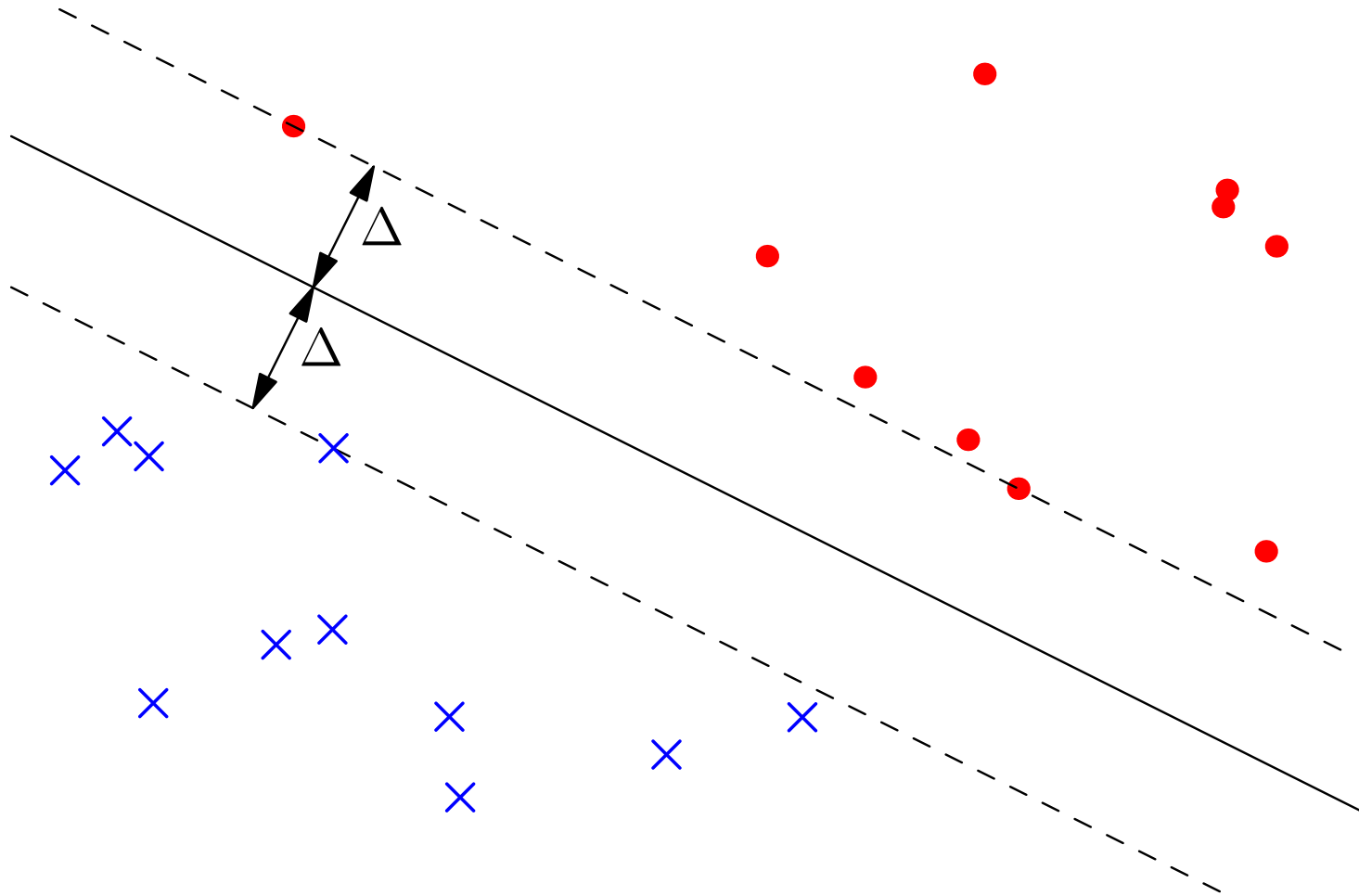
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- SVMs choose the machine with the biggest margins

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- You also need to tune its hyper-parameters (C and sometimes γ)

Lessons

- Machine learning isn't magic
- It works when the learning machine is well attuned to the problem
- Sometimes you can help by preprocessing your data
- Sometimes there is a regularisation term that helps select a simpler machine
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